Supplementary material for “When Does Deep Reinforcement Learning Help Link Adaptation? A Pre-Registered Evaluation Against Calibrated Rules

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How to read this supplement

All files are also provided as CSV or Markdown in the accompanying archive, with SHA-256 hashes in `MANIFEST.sha256`. S1 is the case registry: 67 cases and 108 scheme evaluations of the main round (v9–v17), plus the ledger of the 9 cases and 18 scheme evaluations of the pre-registered robustness checks (76 cases, 126 scheme evaluations in total). S2 gives the per-case comparator tables that exist as result tables (v16, v17 and every robustness stage); for v9–v15 the registry row carries the margin over the best rule, and the per-trace results are in the code and data release. S3 lists every frozen pre-registration with its SHA-256 hash, the commit in which that hash entered `FREEZE.sha256`, the first runner start of the stage (runner heartbeat); all times are UTC+6, the time zone of the run PC and the commit that added its results; the full texts follow. Margins are relative differences in percent, paired per trace; “beats follows rule R1 of the paper (Holm-adjusted Wilcoxon $p < 0.05$ and 90 % CI lower bound above zero).

S1. Case registry

Table 1: Registry of the main round: 67 cases, 108 scheme evaluations (`case_registry.csv`; the `correction` column is in the CSV).

case_id	version	tag	role	rl_scheme	primary_metric	rl_value	best_rule	margin_vs_best_rule_pct	claimA	claimB	win	holm_vs_best	n_seeds	n_traces	status
1	v9	gp_O	primary	Deep A2C v9	goodput	2.8042	Baseline	-8.22			0	4.88e-13	10	100	valid
2	v9	steady_match	secondary	Deep A2C v9	goodput	1.9260	Baseline	-33.13			0	3.61e-17	5	100	valid
3	v9	steady_cold	secondary	Deep A2C v9	goodput	1.9860	Baseline	-31.05			0	3.61e-17	5	100	valid
4	v9	rr_O	robustness	Deep A2C v9	goodput	3.4892	Fixed	-1.57			0	0.000174	10	100	valid
5	v9	eq_O	robustness	Deep A2C v9	goodput	3.3984	Fixed	-5.04			0	1.01e-15	10	100	valid
6	v9	burst6_O	robustness	Deep A2C v9	goodput	2.9071	Baseline	-8.09			0	2.59e-13	10	100	valid
7	v9	burst12_O	robustness	Deep A2C v9	goodput	2.7234	Baseline	-7.60			0	4.63e-13	10	100	valid
8	v9	h0_O	robustness	Deep A2C v9	goodput	2.6838	Baseline	-7.10			0	9.87e-15	10	100	valid
9	v9	h05_O	robustness	Deep A2C v9	goodput	2.7926	Baseline	-8.72			0	8.57e-16	10	100	valid
10	v9	gp_A	ablation	Deep A2C v9	goodput	2.7818	Baseline	-8.95			0	1.2e-14	10	100	valid
11	v9	gp_B	ablation	Deep A2C v9	goodput	2.8226	Baseline	-7.62			0	3.14e-12	10	100	valid
12	v9	pen_O	ablation	Deep A2C v9	goodput	2.7395	Baseline	-10.34			0	7.73e-15	10	100	valid
13	v9	pen_A	ablation	Deep A2C v9	goodput	2.7106	Baseline	-11.28			0	6.41e-16	10	100	valid
14	v9	pen_B	ablation	Deep A2C v9	goodput	2.7730	Baseline	-9.24			0	1.07e-13	10	100	valid
15	v9	orig_O	ablation	Deep A2C v9	goodput	1.9821	Baseline	-35.13			0	3.51e-17	10	100	valid
16	v9	orig_A	ablation	Deep A2C v9	goodput	1.4734	Baseline	-51.78			0	3.51e-17	10	100	valid
17	v9	orig_B	ablation	Deep A2C v9	goodput	1.7548	Baseline	-42.57			0	3.61e-17	10	100	valid
18	v9	orig_O_long1500	ablation	Deep A2C v9	goodput	1.3902	Baseline	-54.50			0	3.51e-17	3	100	valid
19	v10	E2_stale	ablation	A2C	goodput	1.6538	Baseline	-43.12			0	3.51e-17	5	100	valid
20	v10	E3_cellcue	ablation	A2C	goodput	1.5101	Baseline	-48.57			0	3.51e-17	5	100	valid
21	v10	E4_deadline	ablation	A2C	ontime	0.8504	Fixed	0.21			0	0.248	5	100	valid
22	v10	E1_train	ablation	A2C	goodput	1.3800	Baseline	-54.83			0	3.51e-17	5	100	valid
23	v10	E2_stale_nofeat	ablation	A2C	goodput	1.6434	Baseline	-43.47			0	3.51e-17	5	100	valid
24	v10	E3_cellcue_nofeat	ablation	A2C	goodput	1.6095	Baseline	-45.18			0	3.51e-17	5	100	valid
25	v10	E4_deadline_nofeat	ablation	A2C	ontime	0.8474	Fixed	-0.15			0	0.79	5	100	valid
26	v10	E1_train_b	ablation	A2C	goodput	2.9191	Baseline	-4.46			0	4.95e-08	5	100	valid
27	v10	E2_stale_b	ablation	A2C	goodput	2.7493	Baseline	-5.44			0	1.13e-09	5	100	valid
28	v10	E3_cellcue_b	ablation	A2C	goodput	2.7643	Baseline	-5.85			0	4.03e-11	5	100	valid
29	v10	E2_stale_nofeat_b	ablation	A2C	goodput	2.7428	Baseline	-5.66			0	1.41e-09	5	100	valid
30	v10	E3_cellcue_nofeat_b	ablation	A2C	goodput	2.7880	Baseline	-5.04			0	8.97e-09	5	100	valid
31	v10	E5_mixed	ablation	A2C	ontime	0.8013	Context table	-0.27			0	0.526	5	300	valid
32	v10	E4c_deadline	primary	A2C	ontime	0.8514	Fixed	0.35			1	0.00536	5	300	valid
33	v10	E5_mixed_nofeat	ablation	A2C	ontime	0.8107	Context table	0.90			1	0.00114	5	300	valid
34	v11	T1_gp	primary	A2C	goodput	2.9040	Baseline	-2.79			0	7.46e-16	5	300	valid
35	v11	T2_stale	ablation	A2C	goodput	2.7068	Baseline	-5.28			0	3.89e-26	5	300	valid
36	v11	T4_deadline	ablation	A2C	ontime	0.8298	Fixed	-2.19			0	6.83e-17	5	300	valid
37	v11	T1_gp_ppo	primary	Deep A2C v11	goodput	2.9040	Baseline	-2.79			0	7.46e-16	5	300	valid
37	v11	T1_gp_ppo	primary	Deep PPO v11	goodput	2.9191	Baseline	-2.29			0	5.07e-13	5	300	valid
38	v11	T2_stale_ppo	ablation	Deep A2C v11	goodput	2.7068	Baseline	-5.28			0	3.89e-26	5	300	valid
38	v11	T2_stale_ppo	ablation	Deep PPO v11	goodput	2.6838	Baseline	-6.09			0	6.7e-29	5	300	valid
39	v11	T4_deadline_ppo	ablation	Deep A2C v11	ontime	0.8298	Fixed	-2.19			0	6.83e-17	5	300	valid
39	v11	T4_deadline_ppo	ablation	Deep PPO v11	ontime	0.8005	Fixed	-5.65			0	3.27e-37	5	300	valid
40	v11	R2_stale	ablation	Deep A2C v11	goodput	1.4870	Baseline	-47.97			0	5.48e-50	5	300	valid
40	v11	R2_stale	ablation	Deep GRU v11	goodput	2.7236	Baseline	-4.69			0	2.87e-20	5	300	valid
41	v11	R3_cellcue	ablation	Deep A2C v11	goodput	1.4918	Baseline	-48.99			0	5.48e-50	5	300	valid
41	v11	R3_cellcue	ablation	Deep GRU v11	goodput	2.7552	Baseline	-5.79			0	1.59e-33	5	300	valid
42	v11	R2_stale_opp	ablation	Deep A2C v11	goodput	2.6595	Baseline	-6.94			0	7.54e-33	5	300	valid
42	v11	R2_stale_opp	ablation	Deep GRU v11	goodput	2.5102	Baseline	-12.16			0	8.26e-47	5	300	valid
43	v11	R3_cellcue_opp	ablation	Deep A2C v11	goodput	2.7807	Baseline	-4.92			0	8.33e-23	5	300	valid
43	v11	R3_cellcue_opp	ablation	Deep GRU v11	goodput	2.7440	Baseline	-6.17			0	2.46e-34	5	300	valid
44	v12	R1_pf	primary	Deep A2C v12	ontime	0.8421	Fixed	-0.47			0	0.000276	10	500	valid
45	v12	R2_edf	secondary	Deep A2C v12	ontime	0.8717	FixedPU (Oracle)	0.09			0	0.712	5	300	valid
46	v13	X1_pf	primary	Deep A2C v13	ontime	0.8491	Fixed	0.26			1	0.000797	10	500	valid
46	v13	X1_pf	primary	Deep A2C v13 (ensemble)	ontime	0.8472	Fixed	0.04			0	0.135	10	500	valid
47	v13	X1_edf	secondary	Deep A2C v13	ontime	0.8621	Context table	-0.06			0	1	10	500	valid
47	v13	X1_edf	secondary	Deep A2C v13 (ensemble)	ontime	0.8616	Context table	-0.13			0	1	10	500	valid
48	v13	X2_pf_shaped	secondary	Deep A2C v13	ontime	0.8371	Fixed	0.42			1	2.07e-10	10	500	valid
48	v13	X2_pf_shaped	secondary	Deep A2C v13 (ensemble)	ontime	0.8357	Fixed	0.25			1	0.000415	10	500	valid
49	v14	Y0_base	primary	Deep A2C v14	ontime	0.8521	Fixed	0.44	1	1	1	4.75e-07	10	500	valid
49	v14	Y0_base	primary	Deep A2C v14 (ensemble)	ontime	0.8557	Fixed	0.86	1	1	1	4.51e-15	10	500	valid
50	v14	Y0_noshape	secondary	Deep A2C v14	ontime	0.8501	Fixed	0.21	1	1	1	9.66e-05	10	500	valid
50	v14	Y0_noshape	secondary	Deep A2C v14 (ensemble)	ontime	0.8484	Fixed	0.00	1	0	0	1	10	500	valid
51	v14	Y1_lo_tight	robustness	Deep A2C v14	ontime	0.9007	Fixed	0.08	1	0	0	0.658	5	300	valid
51	v14	Y1_lo_tight	robustness	Deep A2C v14 (ensemble)	ontime	0.9057	Fixed	0.64	1	1	1	9.53e-08	5	300	valid
52	v14	Y2_lo_loose	robustness	Deep A2C v14	ontime	0.9389	Context table	-0.77	1	0	0	1.79e-07	5	300	valid
52	v14	Y2_lo_loose	robustness	Deep A2C v14 (ensemble)	ontime	0.9386	Context table	-0.81	1	0	0	3.02e-07	5	300	valid
53	v14	Y3_hi_tight	robustness	Deep A2C v14	ontime	0.6725	Context table	0.08	1	0	0	0.853	5	300	valid
53	v14	Y3_hi_tight	robustness	Deep A2C v14 (ensemble)	ontime	0.6722	Context table	0.04	1	0	0	0.818	5	300	valid
54	v14	Y4_hi_loose	robustness	Deep A2C v14	ontime	0.7296	Fixed	-1.73	1	0	0	3.19e-10	5	300	valid
54	v14	Y4_hi_loose	robustness	Deep A2C v14 (ensemble)	ontime	0.7292	Fixed	-1.79	1	0	0	6.02e-05	5	300	valid
55	v15	Z1_rtt8	primary	Deep A2C v15	ontime	0.5708	Context table	-2.16	1	0	0	2.07e-12	10	500	superseded: PF averaging artefact (v17)
55	v15	Z1_rtt8	primary	Deep A2C v15 (ensemble)	ontime	0.5658	Context table	-3.01	1	0	0	1.27e-17	10	500	superseded: PF averaging artefact (v17)
56	v15	Z2_rtt4	robustness	Deep A2C v15	ontime	0.8019	Fixed	-0.21	1	0	0	1.3e-10	10	500	superseded: PF averaging artefact (v17)
56	v15	Z2_rtt4	robustness	Deep A2C v15 (ensemble)	ontime	0.8036	Fixed	0.00	1	0	0	1	10	500	superseded: PF averaging artefact (v17)

case_id	version	tag	role	rl_scheme	primary_metric	rl_value	best_rule	margin_vs_best_rule_pct	claimA	claimB	win	holm_vs_best	n_seeds	n_traces	status
57	v15	Z3_rtt8_loose	robustness	Deep A2C v15	ontime	0.8475	Fixed	-0.25	1	0	0	9.13e-10	10	500	superseded: PF averaging artefact (v17)
57	v15	Z3_rtt8_loose	robustness	Deep A2C v15 (ensemble)	ontime	0.8495	Fixed	-0.00	1	0	0	1	10	500	superseded: PF averaging artefact (v17)
58	v16	K1_dl_rtt8	primary	Deep A2C v16	ontime	0.5745	Context table	-2.65	1	0	0	1.15e-18	10	500	superseded: PF averaging artefact (v17)
58	v16	K1_dl_rtt8	primary	Deep A2C v16 (ensemble)	ontime	0.5712	Context table	-3.21	1	0	0	3.1e-19	10	500	superseded: PF averaging artefact (v17)
58	v16	K1_dl_rtt8	primary	Deep A2C v16 (offset)	ontime	0.5955	Context table	0.90	1	1	1	0.00103	10	500	superseded: PF averaging artefact (v17)
58	v16	K1_dl_rtt8	primary	Deep PPO v16	ontime	0.5618	Context table	-4.81	1	0	0	1.64e-31	10	500	superseded: PF averaging artefact (v17)
59	v16	K3_gp_rtt8	primary	Deep A2C v16	goodput	4.0724	OLLA	-0.01	0	0	0	1	10	500	superseded: PF averaging artefact (v17)
59	v16	K3_gp_rtt8	primary	Deep A2C v16 (ensemble)	goodput	4.0727	OLLA	0.00	0	0	0	1	10	500	superseded: PF averaging artefact (v17)
59	v16	K3_gp_rtt8	primary	Deep A2C v16 (offset)	goodput	4.0727	OLLA	-0.00	0	0	0	1	10	500	superseded: PF averaging artefact (v17)
59	v16	K3_gp_rtt8	primary	Deep PPO v16	goodput	3.3727	OLLA	-17.19	0	0	0	2.67e-76	10	500	superseded: PF averaging artefact (v17)
60	v16	K2_dl_rtt0	robustness	Deep A2C v16	ontime	0.8360	Fixed	0.28	1	1	1	0.00109	10	500	valid
60	v16	K2_dl_rtt0	robustness	Deep A2C v16 (ensemble)	ontime	0.8350	Fixed	0.15	1	1	1	0.0499	10	500	valid
60	v16	K2_dl_rtt0	robustness	Deep A2C v16 (offset)	ontime	0.8351	Fixed	0.17	1	0	0	0.0502	10	500	valid
60	v16	K2_dl_rtt0	robustness	Deep PPO v16	ontime	0.8079	Fixed	-3.09	0	0	0	2.18e-33	10	500	valid
61	v16	K4_gp_rtt0	robustness	Deep A2C v16	goodput	2.7178	Baseline	-5.52	0	0	0	2.67e-48	10	500	valid
61	v16	K4_gp_rtt0	robustness	Deep A2C v16 (ensemble)	goodput	2.7257	Baseline	-5.25	0	0	0	1.33e-41	10	500	valid
61	v16	K4_gp_rtt0	robustness	Deep A2C v16 (offset)	goodput	2.7250	Baseline	-5.28	0	0	0	1.04e-41	10	500	valid
61	v16	K4_gp_rtt0	robustness	Deep PPO v16	goodput	2.1582	Baseline	-24.98	0	0	0	5.26e-82	10	500	valid
62	v16	L1_dl_rtt8_long	ablation	Deep A2C v16	ontime	0.6068	Context table	1.18	1	1	1	0.00218	5	300	superseded: PF averaging artefact (v17)
62	v16	L1_dl_rtt8_long	ablation	Deep A2C v16 (ensemble)	ontime	0.6100	Context table	1.73	1	1	1	0.000711	5	300	superseded: PF averaging artefact (v17)
63	v16	L3_gp_rtt8_long	ablation	Deep A2C v16	goodput	3.9512	Baseline	-1.10	0	0	0	0.165	5	300	superseded: PF averaging artefact (v17)
63	v16	L3_gp_rtt8_long	ablation	Deep A2C v16 (ensemble)	goodput	3.9576	Baseline	-0.94	0	0	0	0.361	5	300	superseded: PF averaging artefact (v17)
64	v17	M1_dl_rtt8	primary	Deep A2C v17	ontime	0.6126	Context table	1.58	1	1	1	9.06e-13	10	500	valid
64	v17	M1_dl_rtt8	primary	Deep A2C v17 (ensemble)	ontime	0.6306	Context table	4.56	1	1	1	1.62e-43	10	500	valid
64	v17	M1_dl_rtt8	primary	Deep A2C v17 (offset)	ontime	0.5976	Context table	-0.91	1	0	0	8.58e-07	10	500	valid
64	v17	M1_dl_rtt8	primary	Deep A2C v17 (resid-800)	ontime	0.6051	Context table	0.34	1	1	1	0.0244	10	500	valid
65	v17	M2_dl_rtt0	robustness	Deep A2C v17	ontime	0.8617	Context table	0.43	1	1	1	0.0167	10	500	valid
65	v17	M2_dl_rtt0	robustness	Deep A2C v17 (ensemble)	ontime	0.8635	Context table	0.63	1	1	1	5.07e-05	10	500	valid
65	v17	M2_dl_rtt0	robustness	Deep A2C v17 (resid-800)	ontime	0.8532	Context table	-0.57	1	0	0	4.71e-06	10	500	valid
66	v17	M3_gp_rtt8	primary	Deep A2C v17	goodput	3.4679	Fixed	-0.88	0	0	0	0.00723	10	500	valid
66	v17	M3_gp_rtt8	primary	Deep A2C v17 (ensemble)	goodput	3.4679	Fixed	-0.88	0	0	0	0.00723	10	500	valid
66	v17	M3_gp_rtt8	primary	Deep A2C v17 (offset)	goodput	3.4114	Fixed	-2.50	0	0	0	9.86e-12	10	500	valid
67	v17	M4_gp_rtt0	robustness	Deep A2C v17	goodput	3.4378	Fixed	-1.73	0	0	0	1.72e-08	10	500	valid
67	v17	M4_gp_rtt0	robustness	Deep A2C v17 (ensemble)	goodput	3.4746	Fixed	-0.68	0	0	0	0.172	10	500	valid

Table 2: Ledger of the pre-registered robustness checks: 9 cases, 18 scheme evaluations.

stage	description	cases	rows	prereg
W1	Warm-up check (cold vs warm OLLA)	1	2	paper/proposals/W1_warmup_prereg.md
G6	Calibration budget and combined table	1	2	paper/proposals/G6_calibration_budget_prereg.md
NR1	Non-oracle base design ablation	1	3	paper/proposals/NR1_nonoracle_residual_prereg.md
G10	Operating-grid check with the M1 recipe (G2; G3)	2	4	paper/proposals/G10_operating_grid_prereg.md
A6	PF-timing diagnostic	1	1	paper/proposals/A6_pf_timing_prereg.md
NR2	Confirmatory non-oracle replication (Amendment 3)	1	2	paper/proposals/NR2_confirmatory_prereg.md
NR-G10	Operating-grid check with the non-oracle recipe (G2_hi_d10_NR; G3_lo_d18_NR; Amendment 5)	2	4	paper/proposals/NR_G10_operating_grid_prereg.md section 6

S2. Comparator tables

Table 3: v16 cases: residual A2C vs each comparator (trace level) Source: `Optimized/v16/paper/table_core.csv`.

case_tag	scheme	comparator	pct_margin	ci90_lo	ci90_hi	wilcoxon_p	holm_p	beats
K1_dl_rtt8	Deep A2C v16	Baseline	+13.98	13.06	14.90	1.53e-66	1.07e-65	1
K1_dl_rtt8	Deep A2C v16	OLLA	+9.25	8.43	10.06	2.35e-50	1.41e-49	1
K1_dl_rtt8	Deep A2C v16	OLLA (tuned target)	+9.25	8.43	10.06	2.35e-50	1.41e-49	1
K1_dl_rtt8	Deep A2C v16	Multi-OLLA	+8.90	8.10	9.70	1.36e-48	5.45e-48	1
K1_dl_rtt8	Deep A2C v16	Fixed	-1.94	-2.43	-1.44	8.66e-11	1.73e-10	0
K1_dl_rtt8	Deep A2C v16	FixedPU (Oracle)	+0.68	0.49	0.88	1.9e-07	1.9e-07	1
K1_dl_rtt8	Deep A2C v16	Context table	-2.65	-3.13	-2.18	3.83e-19	1.15e-18	0
K1_dl_rtt8	Deep A2C v16	Bandit (global)	+47.85	46.45	49.25	1.26e-83	1.14e-82	1
K1_dl_rtt8	Deep A2C v16	Bandit (delay class)	+48.02	46.62	49.42	1.26e-83	1.14e-82	1
K1_dl_rtt8	Deep A2C v16 (offset)	Baseline	+18.14	17.16	19.13	2.71e-75	1.9e-74	1
K1_dl_rtt8	Deep A2C v16 (offset)	OLLA	+13.24	12.34	14.13	6.15e-64	3.69e-63	1
K1_dl_rtt8	Deep A2C v16 (offset)	OLLA (tuned target)	+13.24	12.34	14.13	6.15e-64	3.69e-63	1
K1_dl_rtt8	Deep A2C v16 (offset)	Multi-OLLA	+12.87	11.99	13.76	2.28e-63	9.13e-63	1
K1_dl_rtt8	Deep A2C v16 (offset)	Fixed	+1.64	1.19	2.09	3.67e-09	7.34e-09	1

case_tag	scheme	comparator	pct_margin	ci90_lo	ci90_hi	wilcoxon_p	holm_p	beats
K1_dl_rtt8	Deep A2C v16 (offset)	FixedPU (Oracle)	+4.36	3.75	4.97	9.17e-26	2.75e-25	1
K1_dl_rtt8	Deep A2C v16 (offset)	Context table	+0.90	0.47	1.33	0.00103	0.00103	1
K1_dl_rtt8	Deep A2C v16 (offset)	Bandit (global)	+53.24	51.77	54.72	1.27e-83	1.14e-82	1
K1_dl_rtt8	Deep A2C v16 (offset)	Bandit (delay class)	+53.43	51.95	54.90	1.27e-83	1.14e-82	1
K1_dl_rtt8	Deep PPO v16	Baseline	+11.46	10.37	12.54	5.51e-45	3.86e-44	1
K1_dl_rtt8	Deep PPO v16	OLLA	+6.83	5.84	7.82	4.31e-24	1.73e-23	1
K1_dl_rtt8	Deep PPO v16	OLLA (tuned target)	+6.83	5.84	7.82	4.31e-24	1.73e-23	1
K1_dl_rtt8	Deep PPO v16	Multi-OLLA	+6.49	5.50	7.48	3.5e-22	7e-22	1
K1_dl_rtt8	Deep PPO v16	Fixed	-4.11	-4.71	-3.51	2.75e-25	1.37e-24	0
K1_dl_rtt8	Deep PPO v16	FixedPU (Oracle)	-1.55	-2.21	-0.89	0.000125	0.000125	0
K1_dl_rtt8	Deep PPO v16	Context table	-4.81	-5.40	-4.21	2.73e-32	1.64e-31	0
K1_dl_rtt8	Deep PPO v16	Bandit (global)	+44.57	43.10	46.04	1.9e-83	1.52e-82	1
K1_dl_rtt8	Deep PPO v16	Bandit (delay class)	+44.74	43.28	46.21	1.69e-83	1.52e-82	1
K3_gp_rtt8	Deep A2C v16	Baseline	+1.80	0.83	2.78	0.0153	0.107	0
K3_gp_rtt8	Deep A2C v16	OLLA	-0.01	-0.03	0.01	0.512	1	0
K3_gp_rtt8	Deep A2C v16	OLLA (tuned target)	+0.58	-0.00	1.17	0.195	1	0
K3_gp_rtt8	Deep A2C v16	Multi-OLLA	+1.44	0.76	2.12	0.000354	0.00283	1
K3_gp_rtt8	Deep A2C v16	Fixed	-0.01	-0.03	0.01	0.512	1	0
K3_gp_rtt8	Deep A2C v16	FixedPU (Oracle)	+6.58	5.42	7.75	3.09e-18	2.78e-17	1
K3_gp_rtt8	Deep A2C v16	Context table	-0.01	-0.03	0.01	0.512	1	0
K3_gp_rtt8	Deep A2C v16	Bandit (global)	-0.01	-0.03	0.01	0.512	1	0
K3_gp_rtt8	Deep A2C v16	Bandit (delay class)	-0.01	-0.03	0.01	0.512	1	0
K3_gp_rtt8	Deep A2C v16 (offset)	Baseline	+1.81	0.83	2.79	0.0162	0.113	0
K3_gp_rtt8	Deep A2C v16 (offset)	OLLA	-0.00	-0.00	0.00	1	1	0
K3_gp_rtt8	Deep A2C v16 (offset)	OLLA (tuned target)	+0.59	0.01	1.18	0.174	1	0
K3_gp_rtt8	Deep A2C v16 (offset)	Multi-OLLA	+1.45	0.77	2.13	0.000331	0.00265	1
K3_gp_rtt8	Deep A2C v16 (offset)	Fixed	-0.00	-0.00	0.00	1	1	0
K3_gp_rtt8	Deep A2C v16 (offset)	FixedPU (Oracle)	+6.59	5.42	7.76	3.45e-18	3.11e-17	1
K3_gp_rtt8	Deep A2C v16 (offset)	Context table	-0.00	-0.00	0.00	1	1	0
K3_gp_rtt8	Deep A2C v16 (offset)	Bandit (global)	-0.00	-0.00	0.00	1	1	0
K3_gp_rtt8	Deep A2C v16 (offset)	Bandit (delay class)	-0.00	-0.00	0.00	1	1	0
K3_gp_rtt8	Deep PPO v16	Baseline	-15.69	-16.65	-14.72	8.5e-74	1.7e-73	0
K3_gp_rtt8	Deep PPO v16	OLLA	-17.19	-18.22	-16.16	3.33e-77	2.67e-76	0
K3_gp_rtt8	Deep PPO v16	OLLA (tuned target)	-16.70	-17.67	-15.72	1.23e-77	1.11e-76	0
K3_gp_rtt8	Deep PPO v16	Multi-OLLA	-15.99	-16.99	-14.99	1.2e-74	3.6e-74	0
K3_gp_rtt8	Deep PPO v16	Fixed	-17.19	-18.22	-16.16	3.33e-77	2.67e-76	0
K3_gp_rtt8	Deep PPO v16	FixedPU (Oracle)	-11.73	-12.75	-10.71	1.22e-53	1.22e-53	0
K3_gp_rtt8	Deep PPO v16	Context table	-17.19	-18.22	-16.16	3.33e-77	2.67e-76	0
K3_gp_rtt8	Deep PPO v16	Bandit (global)	-17.19	-18.22	-16.16	3.33e-77	2.67e-76	0
K3_gp_rtt8	Deep PPO v16	Bandit (delay class)	-17.19	-18.22	-16.16	3.33e-77	2.67e-76	0
K2_dl_rtt0	Deep A2C v16	Baseline	+4.02	3.63	4.40	7.92e-46	5.54e-45	1
K2_dl_rtt0	Deep A2C v16	OLLA	+1.61	1.31	1.91	4.32e-17	1.73e-16	1
K2_dl_rtt0	Deep A2C v16	OLLA (tuned target)	+1.70	1.41	1.99	4.82e-20	2.41e-19	1
K2_dl_rtt0	Deep A2C v16	Multi-OLLA	+1.59	1.32	1.85	4.53e-21	2.72e-20	1
K2_dl_rtt0	Deep A2C v16	Fixed	+0.28	0.14	0.41	0.00109	0.00109	1
K2_dl_rtt0	Deep A2C v16	FixedPU (Oracle)	+0.58	0.36	0.80	2e-05	6.01e-05	1
K2_dl_rtt0	Deep A2C v16	Context table	+0.90	0.60	1.19	4.07e-05	8.14e-05	1
K2_dl_rtt0	Deep A2C v16	Bandit (global)	+12.17	11.59	12.74	1.73e-81	1.39e-80	1
K2_dl_rtt0	Deep A2C v16	Bandit (delay class)	+12.59	12.00	13.19	9.83e-82	8.85e-81	1
K2_dl_rtt0	Deep A2C v16 (offset)	Baseline	+3.90	3.51	4.29	2.51e-43	1.76e-42	1
K2_dl_rtt0	Deep A2C v16 (offset)	OLLA	+1.50	1.18	1.81	5.68e-15	2.27e-14	1
K2_dl_rtt0	Deep A2C v16 (offset)	OLLA (tuned target)	+1.59	1.29	1.90	2.05e-17	1.03e-16	1
K2_dl_rtt0	Deep A2C v16 (offset)	Multi-OLLA	+1.47	1.20	1.75	1.19e-17	7.12e-17	1
K2_dl_rtt0	Deep A2C v16 (offset)	Fixed	+0.17	0.03	0.30	0.0502	0.0502	0
K2_dl_rtt0	Deep A2C v16 (offset)	FixedPU (Oracle)	+0.47	0.24	0.70	0.000307	0.000615	1
K2_dl_rtt0	Deep A2C v16 (offset)	Context table	+0.79	0.50	1.07	0.000114	0.000343	1
K2_dl_rtt0	Deep A2C v16 (offset)	Bandit (global)	+12.04	11.46	12.62	4.81e-81	3.85e-80	1
K2_dl_rtt0	Deep A2C v16 (offset)	Bandit (delay class)	+12.47	11.88	13.07	3.89e-81	3.51e-80	1
K2_dl_rtt0	Deep PPO v16	Baseline	+0.53	0.02	1.03	0.00985	0.00985	1
K2_dl_rtt0	Deep PPO v16	OLLA	-1.80	-2.26	-1.34	1.11e-07	3.34e-07	0
K2_dl_rtt0	Deep PPO v16	OLLA (tuned target)	-1.71	-2.16	-1.26	4.47e-07	8.94e-07	0
K2_dl_rtt0	Deep PPO v16	Multi-OLLA	-1.82	-2.26	-1.39	2.23e-08	8.91e-08	0
K2_dl_rtt0	Deep PPO v16	Fixed	-3.09	-3.45	-2.73	3.63e-34	2.18e-33	0
K2_dl_rtt0	Deep PPO v16	FixedPU (Oracle)	-2.80	-3.14	-2.45	6.15e-32	3.08e-31	0
K2_dl_rtt0	Deep PPO v16	Context table	-2.49	-2.77	-2.21	2.77e-38	1.94e-37	0
K2_dl_rtt0	Deep PPO v16	Bandit (global)	+8.40	7.76	9.04	2.67e-59	2.13e-58	1
K2_dl_rtt0	Deep PPO v16	Bandit (delay class)	+8.82	8.16	9.47	4.01e-61	3.61e-60	1
K4_gp_rtt0	Deep A2C v16	Baseline	-5.52	-6.04	-5.01	2.96e-49	2.67e-48	0
K4_gp_rtt0	Deep A2C v16	OLLA	-0.26	-0.43	-0.09	0.0163	0.0813	0
K4_gp_rtt0	Deep A2C v16	OLLA (tuned target)	-2.47	-2.88	-2.07	8.95e-25	6.27e-24	0
K4_gp_rtt0	Deep A2C v16	Multi-OLLA	+2.35	1.89	2.81	9.09e-19	5.45e-18	1
K4_gp_rtt0	Deep A2C v16	Fixed	-0.26	-0.43	-0.09	0.0163	0.0813	0
K4_gp_rtt0	Deep A2C v16	FixedPU (Oracle)	+3.43	2.95	3.90	4.93e-30	3.95e-29	1
K4_gp_rtt0	Deep A2C v16	Context table	-0.26	-0.43	-0.09	0.0163	0.0813	0
K4_gp_rtt0	Deep A2C v16	Bandit (global)	-0.26	-0.43	-0.09	0.0163	0.0813	0
K4_gp_rtt0	Deep A2C v16	Bandit (delay class)	-0.26	-0.43	-0.09	0.0163	0.0813	0
K4_gp_rtt0	Deep A2C v16 (offset)	Baseline	-5.28	-5.84	-4.71	1.16e-42	1.04e-41	0
K4_gp_rtt0	Deep A2C v16 (offset)	OLLA	+0.00	-0.00	0.01	0.312	1	0
K4_gp_rtt0	Deep A2C v16 (offset)	OLLA (tuned target)	-2.22	-2.67	-1.77	1.92e-18	1.21e-17	0
K4_gp_rtt0	Deep A2C v16 (offset)	Multi-OLLA	+2.62	2.10	3.13	1.73e-18	1.21e-17	1
K4_gp_rtt0	Deep A2C v16 (offset)	Fixed	+0.00	-0.00	0.01	0.312	1	0
K4_gp_rtt0	Deep A2C v16 (offset)	FixedPU (Oracle)	+3.70	3.15	4.25	2.12e-27	1.7e-26	1
K4_gp_rtt0	Deep A2C v16 (offset)	Context table	+0.00	-0.00	0.01	0.312	1	0
K4_gp_rtt0	Deep A2C v16 (offset)	Bandit (global)	+0.00	-0.00	0.01	0.312	1	0

case_tag	scheme	comparator	pct_margin	ci90_lo	ci90_hi	wilcoxon_p	holm_p	beats
K4_gp_rtt0	Deep A2C v16 (offset)	Bandit (delay class)	+0.00	-0.00	0.01	0.312	1	0
K4_gp_rtt0	Deep PPO v16	Baseline	-24.98	-25.82	-24.13	6.58e-83	5.26e-82	0
K4_gp_rtt0	Deep PPO v16	OLLA	-20.80	-21.58	-20.01	7.6e-83	5.32e-82	0
K4_gp_rtt0	Deep PPO v16	OLLA (tuned target)	-22.56	-23.38	-21.73	8.93e-83	5.32e-82	0
K4_gp_rtt0	Deep PPO v16	Multi-OLLA	-18.73	-19.51	-17.94	1.2e-82	5.32e-82	0
K4_gp_rtt0	Deep PPO v16	Fixed	-20.80	-21.58	-20.01	7.6e-83	5.32e-82	0
K4_gp_rtt0	Deep PPO v16	FixedPU (Oracle)	-17.87	-18.51	-17.24	1.51e-83	1.35e-82	0
K4_gp_rtt0	Deep PPO v16	Context table	-20.80	-21.58	-20.01	7.6e-83	5.32e-82	0
K4_gp_rtt0	Deep PPO v16	Bandit (global)	-20.80	-21.58	-20.01	7.6e-83	5.32e-82	0
K4_gp_rtt0	Deep PPO v16	Bandit (delay class)	-20.80	-21.58	-20.01	7.6e-83	5.32e-82	0
L1_dl_rtt8_long	Deep A2C v16	Baseline	+19.29	18.09	20.50	1.01e-48	7.05e-48	1
L1_dl_rtt8_long	Deep A2C v16	OLLA	+14.21	13.11	15.31	1.52e-43	9.1e-43	1
L1_dl_rtt8_long	Deep A2C v16	OLLA (tuned target)	+14.21	13.11	15.31	1.52e-43	9.1e-43	1
L1_dl_rtt8_long	Deep A2C v16	Multi-OLLA	+13.79	12.68	14.90	4.82e-42	1.93e-41	1
L1_dl_rtt8_long	Deep A2C v16	Fixed	+1.74	1.14	2.34	4.48e-06	8.96e-06	1
L1_dl_rtt8_long	Deep A2C v16	FixedPU (Oracle)	+3.90	3.21	4.60	2.04e-16	6.13e-16	1
L1_dl_rtt8_long	Deep A2C v16	Context table	+1.18	0.56	1.81	0.00218	0.00218	1
L1_dl_rtt8_long	Deep A2C v16	Bandit (global)	+55.15	53.10	57.21	6.14e-51	5.53e-50	1
L1_dl_rtt8_long	Deep A2C v16	Bandit (delay class)	+55.15	53.10	57.21	6.14e-51	5.53e-50	1
L3_gp_rtt8_long	Deep A2C v16	Baseline	-1.10	-2.34	0.14	0.0206	0.165	0
L3_gp_rtt8_long	Deep A2C v16	OLLA	-0.16	-0.39	0.07	0.468	1	0
L3_gp_rtt8_long	Deep A2C v16	OLLA (tuned target)	-0.65	-1.48	0.18	0.124	0.87	0
L3_gp_rtt8_long	Deep A2C v16	Multi-OLLA	+0.27	-0.56	1.10	0.532	1	0
L3_gp_rtt8_long	Deep A2C v16	Fixed	-0.16	-0.39	0.07	0.468	1	0
L3_gp_rtt8_long	Deep A2C v16	FixedPU (Oracle)	+5.18	3.68	6.69	2.7e-07	2.43e-06	1
L3_gp_rtt8_long	Deep A2C v16	Context table	-0.16	-0.39	0.07	0.468	1	0
L3_gp_rtt8_long	Deep A2C v16	Bandit (global)	-0.16	-0.39	0.07	0.468	1	0
L3_gp_rtt8_long	Deep A2C v16	Bandit (delay class)	-0.16	-0.39	0.07	0.468	1	0

Table 4: v17 main round (M1–M4): every scheme vs each comparator (trace level) Source: `Optimized/v17/paper/table_core.csv`.

case_tag	scheme	comparator	pct_margin	ci90_lo	ci90_hi	wilcoxon_p	holm_p	beats	jain_rl	jain_cmp	edge_rl	edge_cmp
M1_dl_rtt8	Deep A2C v17	Baseline	+16.62	15.74	17.50	4.04e-77	2.83e-76	1	0.7715	0.7641	0.2148	0.1404
M1_dl_rtt8	Deep A2C v17	OLLA	+11.96	11.21	12.71	2.5e-67	1.25e-66	1	0.7715	0.7802	0.2148	0.1654
M1_dl_rtt8	Deep A2C v17	OLLA (tuned target)	+11.96	11.21	12.71	2.5e-67	1.25e-66	1	0.7715	0.7802	0.2148	0.1654
M1_dl_rtt8	Deep A2C v17	Multi-OLLA	+12.16	11.40	12.92	5.18e-68	3.11e-67	1	0.7715	0.7792	0.2148	0.1658
M1_dl_rtt8	Deep A2C v17	Fixed	+1.87	1.52	2.23	6.74e-17	1.35e-16	1	0.7715	0.7720	0.2148	0.2110
M1_dl_rtt8	Deep A2C v17	FixedPU (Oracle)	+2.70	2.36	3.03	4.28e-32	1.28e-31	1	0.7715	0.7632	0.2148	0.2109
M1_dl_rtt8	Deep A2C v17	Context table	+1.58	1.23	1.94	9.06e-13	9.06e-13	1	0.7715	0.7740	0.2148	0.2122
M1_dl_rtt8	Deep A2C v17	Bandit (global)	+46.55	45.22	47.88	1.46e-83	1.31e-82	1	0.7715	0.6818	0.2148	0.0765
M1_dl_rtt8	Deep A2C v17	Bandit (delay class)	+46.55	45.22	47.88	1.46e-83	1.31e-82	1	0.7715	0.6818	0.2148	0.0765
M1_dl_rtt8	Deep A2C v17 (offset)	Baseline	+13.76	12.93	14.60	1.13e-71	7.88e-71	1	0.7708	0.7641	0.2050	0.1404
M1_dl_rtt8	Deep A2C v17 (offset)	OLLA	+9.21	8.52	9.91	3.08e-58	1.54e-57	1	0.7708	0.7802	0.2050	0.1654
M1_dl_rtt8	Deep A2C v17 (offset)	OLLA (tuned target)	+9.21	8.52	9.91	3.08e-58	1.54e-57	1	0.7708	0.7802	0.2050	0.1654
M1_dl_rtt8	Deep A2C v17 (offset)	Multi-OLLA	+9.40	8.70	10.11	5.88e-59	3.53e-58	1	0.7708	0.7792	0.2050	0.1658
M1_dl_rtt8	Deep A2C v17 (offset)	Fixed	-0.63	-0.91	-0.35	0.000531	0.00106	0	0.7708	0.7720	0.2050	0.2110
M1_dl_rtt8	Deep A2C v17 (offset)	FixedPU (Oracle)	+0.18	-0.18	0.53	0.73	0.73	0	0.7708	0.7632	0.2050	0.2109
M1_dl_rtt8	Deep A2C v17 (offset)	Context table	-0.91	-1.19	-0.63	2.86e-07	8.58e-07	0	0.7708	0.7740	0.2050	0.2122
M1_dl_rtt8	Deep A2C v17 (offset)	Bandit (global)	+42.95	41.65	44.26	1.43e-83	1.28e-82	1	0.7708	0.6818	0.2050	0.0765
M1_dl_rtt8	Deep A2C v17 (offset)	Bandit (delay class)	+42.95	41.65	44.26	1.43e-83	1.28e-82	1	0.7708	0.6818	0.2050	0.0765
M1_dl_rtt8	Deep A2C v17 (resid-800)	Baseline	+15.19	14.32	16.06	2.16e-73	1.51e-72	1	0.7735	0.7641	0.2096	0.1404
M1_dl_rtt8	Deep A2C v17 (resid-800)	OLLA	+10.59	9.85	11.33	1.19e-61	5.96e-61	1	0.7735	0.7802	0.2096	0.1654
M1_dl_rtt8	Deep A2C v17 (resid-800)	OLLA (tuned target)	+10.59	9.85	11.33	1.19e-61	5.96e-61	1	0.7735	0.7802	0.2096	0.1654
M1_dl_rtt8	Deep A2C v17 (resid-800)	Multi-OLLA	+10.78	10.03	11.53	3.13e-62	1.88e-61	1	0.7735	0.7792	0.2096	0.1658
M1_dl_rtt8	Deep A2C v17 (resid-800)	Fixed	+0.62	0.34	0.91	0.00015	0.000301	1	0.7735	0.7720	0.2096	0.2110
M1_dl_rtt8	Deep A2C v17 (resid-800)	FixedPU (Oracle)	+1.44	1.19	1.69	6.33e-18	1.9e-17	1	0.7735	0.7632	0.2096	0.2109
M1_dl_rtt8	Deep A2C v17 (resid-800)	Context table	+0.34	0.05	0.62	0.0244	0.0244	1	0.7735	0.7740	0.2096	0.2122
M1_dl_rtt8	Deep A2C v17 (resid-800)	Bandit (global)	+44.75	43.42	46.09	1.49e-83	1.34e-82	1	0.7735	0.6818	0.2096	0.0765
M1_dl_rtt8	Deep A2C v17 (resid-800)	Bandit (delay class)	+44.75	43.42	46.09	1.49e-83	1.34e-82	1	0.7735	0.6818	0.2096	0.0765
M2_dl_rtt0	Deep A2C v17	Baseline	+4.73	4.40	5.06	3.28e-64	2.3e-63	1	0.9323	0.9235	0.4136	0.3605
M2_dl_rtt0	Deep A2C v17	OLLA	+2.50	2.23	2.77	1.44e-39	8.63e-39	1	0.9323	0.9298	0.4136	0.3889
M2_dl_rtt0	Deep A2C v17	OLLA (tuned target)	+2.50	2.23	2.77	1.44e-39	8.63e-39	1	0.9323	0.9298	0.4136	0.3889
M2_dl_rtt0	Deep A2C v17	Multi-OLLA	+2.30	2.05	2.55	1.68e-39	8.63e-39	1	0.9323	0.9297	0.4136	0.3928
M2_dl_rtt0	Deep A2C v17	Fixed	+0.93	0.72	1.13	5.58e-11	1.12e-10	1	0.9323	0.9262	0.4136	0.4158
M2_dl_rtt0	Deep A2C v17	FixedPU (Oracle)	+1.00	0.80	1.20	3.15e-15	9.44e-15	1	0.9323	0.9288	0.4136	0.4084
M2_dl_rtt0	Deep A2C v17	Context table	+0.43	0.21	0.64	0.0167	0.0167	1	0.9323	0.9299	0.4136	0.4145
M2_dl_rtt0	Deep A2C v17	Bandit (global)	+12.57	12.09	13.06	4.22e-83	3.8e-82	1	0.9323	0.9081	0.4136	0.3053
M2_dl_rtt0	Deep A2C v17	Bandit (delay class)	+12.57	12.09	13.06	4.22e-83	3.8e-82	1	0.9323	0.9081	0.4136	0.3053
M2_dl_rtt0	Deep A2C v17 (resid-800)	Baseline	+3.69	3.36	4.02	2.35e-49	1.64e-48	1	0.9292	0.9235	0.4079	0.3605
M2_dl_rtt0	Deep A2C v17 (resid-800)	OLLA	+1.48	1.22	1.75	2.71e-18	1.62e-17	1	0.9292	0.9298	0.4079	0.3889
M2_dl_rtt0	Deep A2C v17 (resid-800)	OLLA (tuned target)	+1.48	1.22	1.75	2.71e-18	1.62e-17	1	0.9292	0.9298	0.4079	0.3889
M2_dl_rtt0	Deep A2C v17 (resid-800)	Multi-OLLA	+1.28	1.03	1.53	7.8e-17	3.12e-16	1	0.9292	0.9297	0.4079	0.3928
M2_dl_rtt0	Deep A2C v17 (resid-800)	Fixed	-0.07	-0.27	0.12	0.407	0.814	0	0.9292	0.9262	0.4079	0.4158
M2_dl_rtt0	Deep A2C v17 (resid-800)	FixedPU (Oracle)	-0.00	-0.10	0.09	0.85	0.85	0	0.9292	0.9288	0.4079	0.4084
M2_dl_rtt0	Deep A2C v17 (resid-800)	Context table	-0.57	-0.78	-0.36	1.57e-06	4.71e-06	0	0.9292	0.9299	0.4079	0.4145
M2_dl_rtt0	Deep A2C v17 (resid-800)	Bandit (global)	+11.46	10.98	11.93	1.74e-82	1.56e-81	1	0.9292	0.9081	0.4079	0.3053
M2_dl_rtt0	Deep A2C v17 (resid-800)	Bandit (delay class)	+11.46	10.98	11.93	1.74e-82	1.56e-81	1	0.9292	0.9081	0.4079	0.3053
M3_gp_rtt8	Deep A2C v17	Baseline	+8.81	7.93	9.70	4.18e-45	3.76e-44	1	0.5116	0.5173	0.0325	0.0291

case_tag	scheme	comparator	pct_margin	ci90_lo	ci90_hi	wilcoxon_p	holm_p	beats	jain_rl	jain_cmp	edge_rl	edge_cmp
M3_gp_rtt8	Deep A2C v17	OLLA	+1.32	0.67	1.97	0.00121	0.00723	1	0.5116	0.5167	0.0325	0.0314
M3_gp_rtt8	Deep A2C v17	OLLA (tuned target)	+2.14	1.44	2.84	2.42e-06	1.94e-05	1	0.5116	0.5138	0.0325	0.0304
M3_gp_rtt8	Deep A2C v17	Multi-OLLA	+0.20	-0.48	0.87	0.888	1	0	0.5116	0.5138	0.0325	0.0314
M3_gp_rtt8	Deep A2C v17	Fixed	-0.88	-1.44	-0.33	0.00173	0.00723	0	0.5116	0.5056	0.0325	0.0319
M3_gp_rtt8	Deep A2C v17	FixedPU (Oracle)	+0.00	-0.00	0.00	1	1	0	0.5116	0.5116	0.0325	0.0325
M3_gp_rtt8	Deep A2C v17	Context table	-0.88	-1.44	-0.33	0.00173	0.00723	0	0.5116	0.5056	0.0325	0.0319
M3_gp_rtt8	Deep A2C v17	Bandit (global)	+1.32	0.67	1.97	0.00121	0.00723	1	0.5116	0.5167	0.0325	0.0314
M3_gp_rtt8	Deep A2C v17	Bandit (delay class)	+1.70	1.07	2.33	1.44e-05	0.000101	1	0.5116	0.5162	0.0325	0.0312
M3_gp_rtt8	Deep A2C v17 (offset)	Baseline	+7.04	6.33	7.76	5.41e-43	4.87e-42	1	0.5126	0.5173	0.0292	0.0291
M3_gp_rtt8	Deep A2C v17 (offset)	OLLA	-0.33	-0.61	-0.05	0.00873	0.0349	0	0.5126	0.5167	0.0292	0.0314
M3_gp_rtt8	Deep A2C v17 (offset)	OLLA (tuned target)	+0.48	-0.01	0.96	0.136	0.272	0	0.5126	0.5138	0.0292	0.0304
M3_gp_rtt8	Deep A2C v17 (offset)	Multi-OLLA	-1.44	-1.92	-0.95	4.32e-07	2.59e-06	0	0.5126	0.5138	0.0292	0.0314
M3_gp_rtt8	Deep A2C v17 (offset)	Fixed	-2.50	-3.07	-1.92	1.23e-12	9.86e-12	0	0.5126	0.5056	0.0292	0.0319
M3_gp_rtt8	Deep A2C v17 (offset)	FixedPU (Oracle)	-1.63	-2.18	-1.08	1.59e-06	7.97e-06	0	0.5126	0.5116	0.0292	0.0325
M3_gp_rtt8	Deep A2C v17 (offset)	Context table	-2.50	-3.07	-1.92	1.23e-12	9.86e-12	0	0.5126	0.5056	0.0292	0.0319
M3_gp_rtt8	Deep A2C v17 (offset)	Bandit (global)	-0.33	-0.61	-0.05	0.00873	0.0349	0	0.5126	0.5167	0.0292	0.0314
M3_gp_rtt8	Deep A2C v17 (offset)	Bandit (delay class)	+0.05	-0.21	0.30	0.831	0.831	0	0.5126	0.5162	0.0292	0.0312
M4_gp_rtt0	Deep A2C v17	Baseline	+7.07	6.30	7.83	1.62e-39	1.46e-38	1	0.5080	0.5128	0.0275	0.0261
M4_gp_rtt0	Deep A2C v17	OLLA	+0.15	-0.41	0.72	0.963	1	0	0.5080	0.5101	0.0275	0.0271
M4_gp_rtt0	Deep A2C v17	OLLA (tuned target)	-0.90	-1.50	-0.30	0.0164	0.0819	0	0.5080	0.5092	0.0275	0.0274
M4_gp_rtt0	Deep A2C v17	Multi-OLLA	-0.77	-1.35	-0.20	0.0318	0.127	0	0.5080	0.5073	0.0275	0.0286
M4_gp_rtt0	Deep A2C v17	Fixed	-1.73	-2.23	-1.24	2.46e-09	1.72e-08	0	0.5080	0.5072	0.0275	0.0274
M4_gp_rtt0	Deep A2C v17	FixedPU (Oracle)	-1.04	-1.25	-0.83	6.14e-16	4.91e-15	0	0.5080	0.5069	0.0275	0.0275
M4_gp_rtt0	Deep A2C v17	Context table	-1.73	-2.23	-1.24	2.46e-09	1.72e-08	0	0.5080	0.5072	0.0275	0.0274
M4_gp_rtt0	Deep A2C v17	Bandit (global)	+0.15	-0.41	0.72	0.963	1	0	0.5080	0.5101	0.0275	0.0271
M4_gp_rtt0	Deep A2C v17	Bandit (delay class)	+0.51	-0.03	1.04	0.197	0.592	0	0.5080	0.5101	0.0275	0.0269

Table 5: W1: M1 against comparators with cold and warmed-up OLLA Source: `Optimized/v17x/Dataset/table_warmup.csv`.

comparator	cold_margin_pct	cold_ci90_lo	cold_ci90_hi	cold_holm_p	cold_beats	warm_margin_pct	warm_ci90_lo	warm_ci90_hi	warm_holm_p	warm_beats
Baseline	+16.62	15.74	17.50	2.83e-76	1	+16.42	15.52	17.33	4.90e-75	1
OLLA	+11.96	11.21	12.71	1.25e-66	1	+7.49	6.79	8.19	7.05e-46	1
OLLA (tuned target)	+11.96	11.21	12.71	1.25e-66	1	+7.49	6.79	8.19	7.05e-46	1
Multi-OLLA	+12.16	11.40	12.92	3.11e-67	1	+7.71	6.99	8.43	9.22e-46	1
Fixed	+1.87	1.52	2.23	1.35e-16	1	+2.08	1.74	2.42	1.07e-20	1
FixedPU (Oracle)	+2.70	2.36	3.03	1.28e-31	1	+2.87	2.57	3.17	1.36e-38	1
Context table	+1.58	1.23	1.94	9.06e-13	1	+1.75	1.41	2.10	4.82e-15	1
Bandit (global)	+46.55	45.22	47.88	1.31e-82	1	+25.33	24.27	26.38	2.40e-81	1
Bandit (delay class)	+46.55	45.22	47.88	1.31e-82	1	+25.33	24.27	26.38	2.40e-81	1

Table 6: G6: M1 against comparators re-calibrated on 500 traces Source: `Optimized/v17x/Dataset/table_cal500.csv`.

comparator	margin_pct	ci90_lo	ci90_hi	wilcoxon_p	holm_p	beats
Baseline	+16.62	15.74	17.50	4.04e-77	3.23e-76	1
OLLA	+11.96	11.21	12.71	2.50e-67	1.50e-66	1
OLLA-tuned-500	+8.40	7.74	9.07	1.19e-55	5.97e-55	1
Multi-OLLA	+12.16	11.40	12.92	5.18e-68	3.63e-67	1
Fixed-500	+1.87	1.52	2.23	6.74e-17	2.70e-16	1
FixedPU-500	+1.73	1.31	2.14	9.31e-13	2.79e-12	1
Context-500	+1.43	1.08	1.77	3.07e-11	6.13e-11	1
Bandit (global)	+46.55	45.22	47.88	1.46e-83	1.46e-82	1
Bandit (delay class)	+46.55	45.22	47.88	1.46e-83	1.46e-82	1
Context-joint-500	+0.66	0.29	1.03	1.05e-03	1.05e-03	1

Table 7: NR1: residual on the non-oracle base, M1 traces Source: `Optimized/v17x/Dataset/table_nr1.csv`.

comparator	nr1_margin_pct	ci90_lo	ci90_hi	wilcoxon_p	holm_p	beats
Baseline	+18.69	17.81	19.57	1.07e-80	7.52e-80	1
OLLA	+13.95	13.21	14.68	2.40e-76	1.44e-75	1
OLLA (tuned target)	+13.95	13.21	14.68	2.40e-76	1.44e-75	1
Multi-OLLA	+14.15	13.40	14.89	4.51e-76	1.80e-75	1
Fixed	+3.68	3.31	4.05	1.94e-41	3.87e-41	1
FixedPU (Oracle)	+4.52	4.09	4.95	2.95e-46	8.85e-46	1
Context table	+3.38	3.02	3.75	1.51e-37	1.51e-37	1
Bandit (global)	+49.15	47.79	50.51	1.38e-83	1.25e-82	1
Bandit (delay class)	+49.15	47.79	50.51	1.38e-83	1.25e-82	1
M1 (Oracle-base primary)	+1.77	1.55	2.00	2.03e-31	2.03e-31	1

Table 8: NR2: confirmatory replication on fresh traces and seeds Source: `Optimized/v17x/Dataset/table_nr2.csv`.

comparator	margin_pct	ci90_lo	ci90_hi	wilcoxon_p	holm_p	beats
Baseline	+18.19	17.31	19.07	5.828e-79	4.662e-78	1
OLLA	+14.68	13.92	15.45	5.585e-77	3.910e-76	1
OLLA-tuned-500	+10.54	9.94	11.14	3.057e-74	1.528e-73	1
Multi-OLLA	+14.71	13.93	15.49	7.088e-77	4.253e-76	1
Fixed-500	+4.85	4.45	5.24	1.107e-54	4.427e-54	1
FixedPU-500	+4.58	4.17	4.98	3.868e-49	1.160e-48	1
Context-500	+4.33	3.94	4.73	5.943e-48	1.189e-47	1
Context-joint-500	+3.35	2.93	3.76	3.830e-30	3.830e-30	1
M1 (Oracle-base primary)	+2.58	2.32	2.84	5.191e-42	5.191e-42	1

Table 9: G10: M1 recipe at two neighbouring operating points Source: `Optimized/v17x/Dataset/table_cells.csv`.

cell	comparator	margin_pct	ci90_lo	ci90_hi	wilcoxon_p	holm_p	beats
G2_hi_d10	Baseline	+12.40	11.39	13.42	1.92e-41	1.54e-40	1
G2_hi_d10	OLLA	+9.43	8.51	10.36	1.63e-35	9.76e-35	1
G2_hi_d10	OLLA (tuned target)	+6.28	5.50	7.05	2.84e-27	1.42e-26	1
G2_hi_d10	Multi-OLLA	+9.47	8.58	10.35	2.07e-37	1.45e-36	1
G2_hi_d10	Fixed	+1.49	0.99	2.00	4.40e-06	1.76e-05	1
G2_hi_d10	FixedPU (Oracle)	+0.93	0.42	1.43	2.14e-03	6.42e-03	1
G2_hi_d10	Context table	+0.38	-0.11	0.87	2.90e-01	5.80e-01	0
G2_hi_d10	Bandit (global)	+40.63	38.88	42.39	7.21e-51	7.21e-50	1
G2_hi_d10	Bandit (delay class)	+40.63	38.88	42.39	7.21e-51	7.21e-50	1
G2_hi_d10	Context-joint	-0.29	-0.79	0.21	4.59e-01	5.80e-01	0
G3_lo_d18	Baseline	+12.76	12.10	13.42	1.92e-50	1.54e-49	1
G3_lo_d18	OLLA	+9.04	8.46	9.62	3.37e-47	2.02e-46	1
G3_lo_d18	OLLA (tuned target)	+5.94	5.52	6.35	1.89e-45	9.46e-45	1
G3_lo_d18	Multi-OLLA	+8.78	8.25	9.31	3.26e-48	2.28e-47	1
G3_lo_d18	Fixed	+0.59	0.34	0.83	5.49e-03	1.65e-02	1
G3_lo_d18	FixedPU (Oracle)	+0.55	0.31	0.79	3.52e-02	6.97e-02	0
G3_lo_d18	Context table	+0.43	0.19	0.66	3.49e-02	6.97e-02	0
G3_lo_d18	Bandit (global)	+34.56	33.55	35.57	6.08e-51	6.08e-50	1
G3_lo_d18	Bandit (delay class)	+34.56	33.55	35.57	6.08e-51	6.08e-50	1
G3_lo_d18	Context-joint	+0.58	0.29	0.86	7.26e-04	2.90e-03	1

Table 10: NR-G10: non-oracle recipe at the two neighbouring operating points Source: `Optimized/v17x/Dataset/table_cells_nr.csv`.

cell	comparator	margin_pct	ci90_lo	ci90_hi	wilcoxon_p	holm_p	beats
G2_hi_d10	Baseline	+13.90	12.88	14.93	3.54e-44	2.48e-43	1
G2_hi_d10	OLLA	+10.90	9.98	11.82	3.33e-40	1.66e-39	1
G2_hi_d10	OLLA (tuned target)	+7.70	6.96	8.44	1.16e-36	4.62e-36	1
G2_hi_d10	Multi-OLLA	+10.93	10.06	11.80	6.26e-42	3.76e-41	1
G2_hi_d10	Fixed	+2.85	2.32	3.38	5.23e-15	1.57e-14	1
G2_hi_d10	FixedPU (Oracle)	+2.28	1.73	2.82	1.59e-10	3.18e-10	1
G2_hi_d10	Context table	+1.73	1.23	2.22	1.02e-07	1.02e-07	1
G2_hi_d10	Bandit (global)	+42.51	40.74	44.28	7.40e-51	6.66e-50	1
G2_hi_d10	Bandit (delay class)	+42.51	40.74	44.28	7.40e-51	6.66e-50	1
G3_lo_d18	Baseline	+13.42	12.76	14.08	1.11e-50	7.77e-50	1
G3_lo_d18	OLLA	+9.68	9.11	10.24	8.16e-49	4.08e-48	1
G3_lo_d18	OLLA (tuned target)	+6.56	6.15	6.96	3.04e-48	1.22e-47	1
G3_lo_d18	Multi-OLLA	+9.42	8.89	9.94	1.92e-49	1.15e-48	1
G3_lo_d18	Fixed	+1.17	0.90	1.45	3.49e-11	1.05e-10	1
G3_lo_d18	FixedPU (Oracle)	+1.14	0.84	1.44	4.24e-08	4.24e-08	1
G3_lo_d18	Context table	+1.02	0.75	1.28	1.74e-09	3.47e-09	1
G3_lo_d18	Bandit (global)	+35.35	34.34	36.36	6.08e-51	5.47e-50	1
G3_lo_d18	Bandit (delay class)	+35.35	34.34	36.36	6.08e-51	5.47e-50	1

Table 11: NR-G10 vs the G10 per-cell combined table (descriptive, not pre-registered) Source: `paper/analysis/out/nr_g10_ctxj_descriptive.csv`. Columns omitted here are in the CSV.

cell	n_traces	n_seeds	ctxj_mean	primary_mean	margin_pct	ci90_lo	ci90_hi	wilcoxon_p	hier_ci95_lo	hier_ci95_hi	seeds_above
G2_hi_d10	300	5	0.5422	0.5478	1.04	0.49	1.59	1.92e-03	0.14	2.08	5
G3_lo_d18	300	5	0.8993	0.9098	1.17	0.87	1.47	2.82e-10	0.75	1.6	5

Table 12: A6: PF-averaging definition and multi-process HARQ (rules only) Source: `Optimized/v17x/Dataset/a6_pf_timing.csv`. Columns omitted here are in the CSV.

cell	pf_mode	rtt	nproc	scheme	goodput_mean	goodput_ci_lo	goodput_ci_hi	jain_mean	edge5_mean
acked_rtt0_np1	acked	0	1	Baseline	2.8952	2.6802	3.1101	0.5302	0.0141
acked_rtt0_np1	acked	0	1	OLLA	2.8710	2.6589	3.0832	0.5317	0.0150
acked_rtt0_np1	acked	0	1	Fixed	2.6672	2.4636	2.8707	0.5218	0.0177
acked_rtt8_np1	acked	8	1	Baseline	2.9238	2.6696	3.1780	0.5336	0.0254
acked_rtt8_np1	acked	8	1	OLLA	2.9450	2.6906	3.1994	0.5342	0.0262
acked_rtt8_np1	acked	8	1	Fixed	2.9992	2.7372	3.2612	0.5392	0.0267
acked_rtt8_np8	acked	8	8	Baseline	4.2307	3.8741	4.5873	0.4361	0.0015
acked_rtt8_np8	acked	8	8	OLLA	4.2208	3.8610	4.5806	0.4360	0.0011
acked_rtt8_np8	acked	8	8	Fixed	4.2583	3.8843	4.6324	0.4374	0.0017
scheduled_rtt0_np1	scheduled	0	1	Baseline	3.2399	2.9434	3.5364	0.5004	0.0199
scheduled_rtt0_np1	scheduled	0	1	OLLA	3.2546	2.9592	3.5499	0.5010	0.0194
scheduled_rtt0_np1	scheduled	0	1	Fixed	3.4748	3.1689	3.7808	0.4996	0.0196
scheduled_rtt8_np1	scheduled	8	1	Baseline	3.2881	2.9870	3.5893	0.4936	0.0192
scheduled_rtt8_np1	scheduled	8	1	OLLA	3.2796	2.9793	3.5799	0.4929	0.0191
scheduled_rtt8_np1	scheduled	8	1	Fixed	3.4115	3.1104	3.7127	0.4971	0.0225
scheduled_rtt8_np8	scheduled	8	8	Baseline	3.2098	2.9078	3.5118	0.4977	0.0211
scheduled_rtt8_np8	scheduled	8	8	OLLA	3.2011	2.9046	3.4977	0.4965	0.0215
scheduled_rtt8_np8	scheduled	8	8	Fixed	3.4360	3.1406	3.7314	0.4979	0.0197

Table 13: A3c: per-TTI decision time (one laptop core; descriptive) Source: `Optimized/v17x/Dataset/table_a3c.csv`.

users	scheme	runtime	median_us	p99_us	p99_9_us	max_us
10	Noise floor (empty loop)	Python	0.10	0.30	0.50	31.50
10	Fixed AMC	NumPy	12.30	37.70	58.80	1091.00
10	Context-joint	NumPy	31.90	92.10	150.40	1127.10
10	OLLA	NumPy	33.00	91.40	177.40	1240.40
10	NR2 (State Construction)	NumPy	106.30	263.00	402.40	2656.30
10	NR2 (Forward Pass)	NumPy	69.90	190.60	244.70	1709.00
10	NR2 (Forward Pass)	onnxruntime	68.00	175.70	268.00	2606.80
10	NR2 (Full Policy Step)	NumPy	180.40	413.80	632.60	2290.70
10	NR2 (Full Policy Step)	onnxruntime	186.10	464.90	679.50	2623.20
50	Noise floor (empty loop)	Python	0.10	0.30	0.50	31.50
50	Fixed AMC	NumPy	11.50	28.80	48.20	1073.10
50	Context-joint	NumPy	39.10	122.60	163.30	1158.90
50	OLLA	NumPy	33.70	88.80	173.50	2428.90
50	NR2 (State Construction)	NumPy	112.20	237.70	332.50	1884.30
50	NR2 (Forward Pass)	NumPy	124.60	289.50	382.00	1456.00
50	NR2 (Forward Pass)	onnxruntime	97.50	234.40	426.60	11884.70
50	NR2 (Full Policy Step)	NumPy	255.30	669.40	973.10	3027.30
50	NR2 (Full Policy Step)	onnxruntime	237.80	695.60	927.60	2160.20
100	Noise floor (empty loop)	Python	0.10	0.30	0.50	31.50
100	Fixed AMC	NumPy	13.10	46.90	106.80	1067.50
100	Context-joint	NumPy	35.00	136.20	183.10	1845.50
100	OLLA	NumPy	37.20	97.00	157.20	2591.20
100	NR2 (State Construction)	NumPy	124.20	276.90	473.90	2703.30
100	NR2 (Forward Pass)	NumPy	164.60	429.80	561.60	1625.80
100	NR2 (Forward Pass)	onnxruntime	136.25	298.90	410.30	11805.50
100	NR2 (Full Policy Step)	NumPy	297.90	724.20	1062.11	11999.20
100	NR2 (Full Policy Step)	onnxruntime	284.90	898.90	1204.70	3722.60

S3. Pre-registrations, hashes and freeze timeline

Table 14: Every frozen file verifies against `FREEZE.sha256`. For each stage the hash entered the freeze file before the first runner start and before the results commit. Full hashes are in `freeze_timeline.csv`.

file	stage	verifies	freeze_commit	freeze_time_utc6	first_runner_start_utc6	results_commit	results_time_utc6
G6_calibration_budget_prereg.md	G6	yes	8ab7be1	2026-11-04 22:42	2026-11-05 16:48:42	0386055	2026-11-08 05:39
W1_warmup_prereg.md	W1	yes	ebce975	2026-11-04 21:44	2026-11-05 16:48:42	0386055	2026-11-08 05:39
NR1_nonoracle_residual_prereg.md	NR1	yes	ebce975	2026-11-04 21:44	2026-11-05 16:48:42	0386055	2026-11-08 05:39
G10_operating_grid_prereg.md	G10	yes	d08807f	2026-11-05 16:45	2026-11-08 09:44:26	fd6d7fb	2026-11-10 15:35
A6_pf_timing_prereg.md	A6	yes	8ab7be1	2026-11-04 22:42	2026-11-05 16:48:42	0386055	2026-11-08 05:39
common_rules.md		yes	8ab7be1	2026-11-04 22:42			

file	stage	verifies	freeze_commit	freeze_time_utc6	first_runner_start_utc6	results_commit	results_time_utc6
checkpoint5_list.md		yes	d08807f	2026-11-05 16:45			
compute_basis.md		yes	ebce975	2026-11-04 21:44			
NR2_confirmatory_prereg.md	NR2	yes	a12d322	2026-11-08 09:09	2026-11-10 03:52:44	fd6d7fb	2026-11-10 15:35
nr2_stats.py		yes	a12d322	2026-11-08 09:09			
AMENDMENTS.md		yes	d7fd804	2026-11-10 15:37			
NR_G10_operating_grid_prereg.md	NR-G10	yes	d7fd804	2026-11-10 15:37	2026-11-10 15:53:31	97fc571	2026-11-11 06:18

G6_calibration_budget_prereg.md

SHA-256: `c629332df0eb98086d201a61505758716bfde5dbfc3dd33175d1d24bad5b8bbd`

Pre-registration G6: comparator calibration budget and the combined comparator (v17b-cal)

STATUS: FROZEN (Checkpoint 5, 4 Nov 2026), amended once at Checkpoint 5b (4 Nov 2026) before any run started (`AMENDMENTS.md`, amendment 1: composite defined in full; profile fixed to `full`; report point after `g6a`). §§1–6 do not change after the freeze-v2 commit recorded in `paper/STATUS.md`. Results are appended in §8 only. Common rules: `common_rules.md`. **Class:** essential. **No RL training**: the RL per-trace values come from the stored M1 results.

1. Questions

- **Q1 (budget):** does M1's Claim B survive when Fixed, FixedPU, the Context table and tuned-target OLLA are re-calibrated with a budget comparable to RL's, **using searches that cannot stop at a local optimum where that is feasible**?
- **Q2 (information):** does M1 beat a calibrated rule that combines the learner's cues (slack × class × HARQ-process state)?

What the original calibration did (code read 4 Nov 2026), which is why G6 exists:

Rule	Original calibration (v17)
Fixed	9 global offsets, 20 traces (`calibrateFixedClass_v17.m`): exhaustive
FixedPU	the **same 9 runs**; each class takes its own argmax from per-class metrics. The finished 6-entry table is never evaluated jointly (separable grid)
Context table	**coordinate ascent**, 2 passes from all-zero, 10 traces (`calibrateCtx_v17.m`): can stop at a local optimum
OLLA (tuned target)	6 targets, 10 traces: exhaustive
Residual A2C	3000 episodes × 10 seeds

2. Design

2.1 Fixed parts

- **RL under test (frozen):** M1 primary residual A2C (10 seeds) and its ensemble (secondary). Per-trace values come from `results_v17_M1_dl_rtt8.mat` (`metrics.primary`). REPRO: re-evaluating seed 1 on traces 1–20 must reproduce the stored values exactly.
- **Calibration traces:** 500 fresh traces, seeds **600001–600500**, CRN draws **610001+**. The code asserts disjointness from every `trace_seed_base` / `crn_seed_base` range of v9–v17 and from 60000+ and 80000+ (validation).
- **Evaluation:** the original M1 traces 500001–500500 with CRN 510001+.
- **Calibration objective** (all rules): the primary metric, `sum(okU)/sum(resU)` over users and calibration traces after burn-in, exactly as `calibrateCtx_v17.m: score`. Ties are broken by the smallest Σ|offset|, then lexicographically.
- **Environment:** M1's (RTT 8, 8 processes, `pf_avg = 'scheduled'`, drain, useful bits, cold-start OLLA).

2.2 Single-cue rules, re-calibrated on the 500 traces ("-500") No coordinate ascent is used for Fixed, Context or OLLA. 1. **Fixed-500:** exhaustive over offsets −5…+3 (9 candidates). 2. **FixedPU-500 (class oracle):** - **(a) separable grid**, as in v17: the 9 global-offset runs; each of the 6 classes takes the argmax of its per-class metric; - **(b) joint refinement:** coordinate ascent on the **joint** 6-entry table, evaluated as a whole on the 500 traces. Sweeps over classes in lexicographic order; each class tries the 8 other offsets in −5…+3; a move is accepted if the joint objective rises by more than 1e-6; at most 3 sweeps. - FixedPU-500 is the table with the higher calibration objective (never lower than (a)). This only **strengthens** FixedPU relative to v17. The paper notes that v17's FixedPU was separable. 3. **Context-500: exhaustive** over all $7^3 = 343$ tables (3 slack bins, edges [2 5], offsets −4…+2). 4. **OLLA-tuned-500:** exhaustive over targets {0.01, 0.02, 0.03, 0.05, 0.07, 0.10, 0.15, 0.20, 0.30, 0.50} × down-steps {0.25, 0.5, 1.0} dB (30 settings; also answers attack #19 / G9).

2.3 Combined comparator, "Context-joint-500"

- **Cells:** slack ≤ 2 / 3–5 / > 5 TTI × delay class d ≤ 4 / d ≥ 5 × speed class < 10 / 10–20 / > 20 m/s (`userClass_v17`; true values, as for FixedPU) × busy HARQ processes 0 / ≥ 1 (`sum(H.proc_busy(u,:)) >= 1` at the decision). **3 × 2 × 3 × 2 = 36 cells.** Busy ≥ 3 merges into ≥ 1 (1.3 % of M1 decisions, `table_action_by_state.csv`).
- **Offsets:** −4…+2 CQI steps on the OLLA reference; applied to new transmissions.
- **Composite table (deterministic; defined here, not only in the code).** Let a cell be `(b, d, v, u)` with slack bin `b ∈ {1,2,3}` (≤ 2, 3–5, > 5 TTI), delay class `d ∈ {1,2}`, speed class `v ∈ {1,2,3}` and busy bin `u ∈ {0,1}`. With the re-calibrated tables `F` (Fixed-500 offset, a scalar), `P(d,v)` (FixedPU-500, the joint-refined 2×3 table) and `X(b)` (Context-500, the 3-entry table): `composite(b,d,v,u) = min(2, max(−4, P(d,v) + X(b) − F))`, with all quantities integer CQI steps and no rounding. It does not depend on `u`. It is evaluated once after `g6a` and written to `composite_table.csv` before `g6b` starts. It is the **initial table of start S0**, the **fallback value of every sparse cell**, and one of the coarser tables in the sanity check. No other "coarser table" exists in this pre-registration.
- **Sparse cells:** one counting pass with the composite table counts scheduled new-transmission decisions per cell on the 500 calibration traces. A cell with **fewer than N = 200** decisions is **not searched** and keeps its composite value ("fallback": the value defined above). The counts and fallback list are written to `ctxj_cells.csv`.
- **Search (coordinate ascent; exhaustive search over 7^{36} tables is infeasible):**
  - **4 starts:** S0 = the composite table; R1–R3 = random initial offsets, uniform on −4…+2 per searched cell, with generator seeds 90017 + k.
  - One **sweep** visits the searched cells in lexicographic order. Each cell tries the 6 other offsets and accepts the best improvement over 1e-6 (evaluated on all 500 traces).
  - **Every start continues until a full sweep makes no accepted move** ("the calibration objective stops improving"). Safety cap: 6 sweeps per start. A start that hits the cap is reported as such.
  - **Final table:** the start with the highest calibration objective (ties → S0).
  - Checkpoint after every sweep.
- **Sanity check, before any evaluation:** the final table's calibration objective must be ≥ that of **each** coarser table: Fixed-500, FixedPU-500, Context-500 and the composite table. If it is not, it is reported as a **search failure**, and `g6c` refuses to run. Because S0 starts at the composite and a move is accepted only if the objective rises, this can fail only through a bug.
- **Freeze rule:** all tables are written to `cal500_tables.txt` and hashed (SHA-256, logged) **before** any evaluation-trace run.
- **Overfitting visibility:** for every re-calibrated table the calibration-set objective and the evaluation-set on-time ratio are reported side by side (`table_cal500.csv`), for all five rules and for the single-cue originals.

2.4 Profiles (decided at Checkpoint 5b, before any result): G6 runs the `full` profile Both profiles stay defined so that the wording and G10 can refer to them.

Profile	Single-cue rules	Combined search	What differs from `full`
full (chosen for G6)	as §2.2	as §2.3: 4 starts, every sweep evaluated on all 500 calibration traces, every start run to convergence	—
lite (`G6B_PROFILE=lite`; **not used**)	as §2.2	the 4 starts run to convergence on the **first 100** calibration traces only; the best **two** final tables are then polished on all 500 traces (sweeps until no accepted move, cap 6); the final table is the better of the two by the 500-trace objective	(i) **100 instead of 500 calibration traces per candidate during the restart phase**; (ii) only two of the four starts are ever evaluated on the full budget; restarts and the cap of 6 sweeps are unchanged; the grid of offsets and the 36 cells are unchanged

`lite` is not used in this study. It is kept only because it is defined in `G10`, which explains why G10 also uses `full`.

2.5 Report point after `g6a` (descriptive only) When `g6a` is done, the runner writes `REPORT_g6a.txt` and continues without waiting. It contains **calibration-set information only** (no evaluation trace has been touched): per rule the calibration objective of the v17 table and of the re-calibrated table; the selected tables (Fixed-500, FixedPU-500 separable and joint, Context-500, OLLA-tuned-500); whether the joint refinement changed FixedPU; how the exhaustive Context table differs from the v17 table; the composite table; and the projected time of `g6b`. Nothing in `g6b`, `g6c` or any decision rule depends on this report; the author may read it, and it **cannot** change any pre-registered choice.

3. Decision rules (pre-registered)

- **Family:** {Baseline, OLLA, OLLA-tuned-500, Multi-OLLA, Fixed-500, FixedPU-500, Context-500, Bandit (global), Bandit (delay class), **Context-joint-500**}: **10 comparators, Holm over 10.**
- **"Beats X":** two-sided Wilcoxon signed-rank on paired per-trace values, Holm-adjusted $p < 0.05$, and the 90 % CI lower bound of the % margin > 0 (`common_rules.md` R1).
- **Claim A** = beats OLLA, OLLA-tuned-500, Multi-OLLA. **Claim B-500** = beats Fixed-500, FixedPU-500, Context-500. **Joint parity (JP)** = beats Context-joint-500.
- **Seed-level rule** (`common_rules.md` R2): for the best comparator in each claim the hierarchical seed × trace 95 % CI lower bound is reported. If a claim passes at trace level but this lower bound is ≤ 0, the paper states the claim with "not robust to seed resampling".

4. Pre-registered wording (also governs contribution C2)

Outcome	Paper statement
A, B-500 and JP pass (seed-level lower bounds > 0)	C2 keeps "beats **all calibrated comparators**". The paper may use **"same-information"**: "With rules re-calibrated by exhaustive search on 500 fresh traces, and a table that combines deadline slack, user class and HARQ-process state, residual A2C still beat every comparator (+x % vs the best rule)."
A, B-500 and JP pass, but a seed-level lower bound ≤ 0	as above, followed by "; the margin over [comparator] is not robust to seed resampling (hierarchical 95 % CI [a, b] %)"
A and B-500 pass, JP fails	C2 becomes "beats **the OLLA family and the single-cue calibrated rules**". The combined table is reported as matching RL (or beating it, if significant): "A calibrated rule combining the learner's cues matched the learned policy (margin x %, CI …)." "Same-information" is not used
A passes, B-500 fails	The headline weakens to Claim A: "Residual A2C beats the OLLA family under realistic HARQ timing (+x %); against calibrated rules its advantage depends on their calibration budget."
A fails	"The deadline gain did not survive re-calibration." The paper becomes a negative and methodological result
Sanity check fails (search failure)	No wording is decided. The author is told before any evaluation; the search is re-run after the bug is fixed (a new, dated pre-registration note)

5. Compute (basis `compute_basis.md`: 0.095 s per 200-TTI evaluation at RTT 8 with 2 workers, an upper bound; the preflight measures the deterministic-rule rate, which should be lower because no network is evaluated)

Step	Evaluations	Wall time
Fixed-500 and FixedPU-500 (a): shared 9 offsets × 500	4,500	0.1 h
FixedPU-500 (b): ≤ 3 sweeps × 48 tables × 500	≤ 72,000	≤ 1.9 h

Step	Evaluations	Wall time
Context-500 exhaustive: 343 × 500	171,500	4.5 h
OLLA-tuned-500: 30 × 500	15,000	0.4 h
`g6a` total	**263,000**	**6.9 h**
Combined: one sweep ≤ 36 cells × 6 × 500	108,000	2.85 h per sweep
`g6b` minimum (4 starts × 2 sweeps) / **expected** (× 3) / **cap** (× 6)	0.86 M / 1.30 M / 2.59 M	**22.8 h / 34.2 h / 68.4 h**
`g6c`: counting pass + evaluation of the re-calibrated rules + REPRO	≈ 8,000	0.2 h
G6 total (full)		**≈ 30 h min, ≈ 41 h expected, ≤ 76 h cap**
G6 total (lite, not used)	`g6b` ≈ 0.26 M (restarts on 100 traces) + ≈ 0.43 M (polish)	*≈ 25 h*

Sweeps skip cells below the 200-decision threshold, so real sweeps are cheaper than the ≤ 36-cell bound. If the preflight shows a measured rate below the basis, all times scale down proportionally. The runner prints the projected time and pauses if the measured rate is more than 1.5× the basis (`common_rules.md` R5).

6. Multiplicity Adds **1 case** (M1-cal500) with 2 scheme rows (primary, ensemble) to the registry. It is a robustness check of the primary claim and cannot create a new claim.

7. Stage specification for the runner (`Optimized/v17x/`) Stages `g6a`, `g6b`, `g6c`, implemented by `stage_g6a(ctx)`, `stage_g6b(ctx)`, `stage_g6c(ctx)` (interface in `Optimized/v17x/README_v17x_runner.md`). `g6a` also provides `stage_g6a_report(ctx)` (§2.5). Run order in the programme: G1 → W1 → NR1 → **G6** → G10 → A1/A3 → A6. - **`g6a`** writes `composite_table.csv` and `REPORT_g6a.txt`; units: `sep_grid_chunkK` (25 traces each), `fixedpu_joint_sweepS`, `ctx_exh_chunkK` (7 chunks of 49 tables), `olla_set_K` (30). Writes `cal500_single.txt`. - **`g6b`** units: `start_k_sweep_s` (checkpoint after each sweep); writes `ctxj_cells.csv` and `cal500_tables.txt` with the SHA-256 and the sanity check. - **`g6c`** refuses to run if the sanity check failed or the hash differs. Evaluates on traces 500001–500500, runs REPRO, computes REPLCMP/REPL/REPLSEED for the 10-comparator family and the hierarchical seed-level CIs, writes `fill_summary_v17b_cal.txt`, `table_cal500.csv` and the verdict row of §4.

8. Results *(empty until run)*

W1_warmup_prereg.md

SHA-256: `3a69b7d023c601f8c221ca03b4ec60b67d2953c2855c23878683ddf8330f1b5d`

Pre-registration: W1, warm-up robustness of the M1 comparison (v17w)

STATUS: FROZEN (Checkpoint 5, 4 Nov 2026). §§1–5 do not change after the freeze commit recorded in `paper/STATUS.md`; results go in §7. Common rules: `common_rules.md`. **Class: essential. No retraining**: stored policies and stored calibrated tables are used.

1. Question In v17, every scheme starts each 200-TTI trace with a cold OLLA loop (`olla_warm_ttis = 0`). Scoring begins after a 50-TTI burn-in. Does the size of the gain over OLLA (+11.96 %, R06), or the result against calibrated rules (Claim B), depend on that cold start?

2. Design

- **Traces:** M1 evaluation traces 500001–500500 with CRN 510001+, as in M1.
- **Warm-up:** `env.olla_warm_ttis = 200`.
  - The frozen evaluator (`evalPolicy_v17.m` L30–60) prepends 200 **unscored** TTIs from an independent channel segment of the same users. The segment's outcome draws are common to all schemes on a trace.
  - Each scheme runs its **own** policy during the warm-up: OLLA, Fixed, FixedPU, Context and the bandits update their loops; the A2C policies act with their own state.
  - Returned metrics exclude the warm-up (L673–676). On-time counting starts after warm-up + burn-in (`trafficInit_v17`).
- **Known exception (pre-stated):** the frozen code applies the warm-up only to schemes with an outer loop (`useOLLA`), so **Baseline is evaluated without warm-up**. Baseline is open-loop, so this does not change what it is. It is outside Claims A and B and is footnoted in the table. The frozen code is not changed.
- **Schemes:** the 9 comparators, the M1 primary (10 seeds) and the ensemble. Offset-800 and resid-800 are not re-run.
- **Calibrated tables:** the stored M1 tables (`results.env`) form the **primary** analysis. A secondary analysis re-calibrates Fixed/FixedPU/Context on 20 calibration traces with `calib_warm = true` (cheap) and reports them alongside.

3. Pre-registered rules

- **The primary claim of the paper stays the cold-start Claim B** (vs calibrated rules, as in v17). W1 is a robustness check of it and of the size of the gain over OLLA.
- **"Beats" machinery:** `common_rules.md` R1 (Holm over 9) and the seed-level rule R2.
- **Gain over OLLA (defined here):** for each condition (cold, warm) the margin of the primary scheme over the **best-performing member of the OLLA family** {OLLA, OLLA (tuned target), Multi-OLLA} in that condition, with its 90 % CI. In cold M1 this is OLLA (+11.96 %).
- **Abstract rule (changed at Checkpoint 5):** the Results section reports **both** the cold-start and the warm-up numbers. **The abstract and introduction quote the smaller of the two gains over the OLLA family, whichever it is**, labelled as such. No overlap test is used.
- **Claim B under warm-up:**
  - *passes* → the text says the result is robust to a 200-TTI warm-up;
  - *fails* → C2 is qualified in the abstract ("with loops started at the beginning of each trace"), and the warm-up result is stated in Results and Limitations.
- **Baseline footnote (table T4/F6):** Baseline is open-loop with no adaptive state; the frozen evaluator applies the warm-up only to schemes with an outer loop, so Baseline's warm and cold values are the same system. This is footnoted, not changed in the code.

4. Compute (basis 0.095 s per 200-TTI evaluation, so ≈ 0.19 s per 400-TTI run)

- Runs: (9 comparators, of which 2 bandits × 10 seeds, so 7 + 20 = 27) + 10 A2C + 1 ensemble = **38 × 500 = 19,000** runs.
- Time: **≈ 1.0 h**, plus about 10 min for the secondary re-calibration and a preflight.

5. Multiplicity This adds 1 case (M1-warm200) with 2 rows (`common_rules.md` R7).

6. Stage specification for the runner (`Optimized/v17x/`, stage `w1`) Implemented as `stage_w1(ctx)`. Units: `comparators_det`, `bandit_seed_K` (10), `a2c_seed_K` (10), `ens`, `repro20`, `secondary_recal`. Outputs `fill_summary_v17w.txt` and `table_warmup.csv`. The same steps as the original prompt below apply.

6b. Original implementation prompt (kept for reference)

> In `Optimized/v17/`, implement W1 exactly as in `paper/proposals/W1_warmup_prereg.md` (frozen). 1. Write `w1/w1_eval_warmup.m`. Load `Dataset/results_v17_M1_dl_rtt8.mat` (`env`, `policies`, scheme list). Set `env.olla_warm_ttis = 200`; keep everything else. Regenerate M1's evaluation traces and CRN draws as `runComparison_v17.m` L183–191 does. Evaluate the 9 comparators (same ptypes and policy sources as `runComparison_v17.m` L206–232 and L300–316; bandits for all 10 seeds), `Deep A2C v17` (10 seeds) and `Deep A2C v17 (ensemble)` with the frozen `evalPolicy_v17.m`. Compute per-trace primary with `userMetric_v17` and goodput, Jain and edge5 as in `runComparison_v17.m`. 2. Sanity check: with `olla_warm_ttis = 0`, re-evaluate 20 traces and require bit-identity with stored `metrics.primary` for every scheme. Stop if not. 3. Compute REPLCMP/REPL/REPLSEED with the functions of `analyze_v17_replication.m` (Holm over 9). Write `Dataset/fill_summary_v17w.txt` and `paper/table_warmup.csv`, with cold and warm margins side by side (cold = stored M1). 4. Secondary: re-calibrate Fixed, FixedPU and Context with `env.calib_warm = true` on 20 calibration traces (`calibrateFixedClass_v17`, `calibrateCtx_v17`), re-evaluate those three under warm-up, and report them in a separate table. 5. Write `README_v17w.md` with the verdict per §3. Note that Baseline has no warm-up in the frozen code. Branch `v17w`.

7. Results *(empty until run)*

NR1_nonoracle_residual_prereg.md

SHA-256: `ed4e5e38faadf9ee7ecff10fa0c65367413aaa91526efec9c720d780716f0258`

Pre-registration: NR1, residual on a non-oracle base under realistic HARQ (v17n)

STATUS: FROZEN (Checkpoint 5, 4 Nov 2026). §§1–6 do not change after the freeze commit recorded in `paper/STATUS.md`; results go in §8. Common rules: `common_rules.md`. **Class: recommended (approved).** It is the only way to remove limitation #25 (oracle-class inheritance) for M1.

1. Background (campaign check, 4 Nov 2026)

- The base rule is selected automatically on 20 calibration traces as the better of {Fixed, FixedPU} (`selectBaseRule_v17.m` L36–50).
- **The non-oracle Fixed base was used** in v13 X1/X1_edf/X2, v14 Y0/Y0_noshape/Y1–Y3, v15 Z2/Z3 and v16 K2–K4.
- The **valid WINs with a non-oracle base** are v13 X2, v14 Y0_base, v14 Y0_noshape and v16 K2. All are **RTT-0 deadline cases under the delivered-bits PF variant** (S08a, S08b).
- **At RTT 8 under v17, M1 used the FixedPU (class-oracle) base; there is no non-oracle run.** v16 K1/L1 (fixed_class, RTT 8) are superseded.
- These are cited in the paper as partial evidence (reviewer_attacks #25).

2. Question Does the M1 result survive when the residual policy acts on the **non-oracle Fixed** base (one global offset) instead of the FixedPU class oracle?

3. Design

- **Configuration:** identical to M1 (RTT 8, 8 processes, `pf_avg = 'scheduled'`, drain, useful bits, 3000 episodes, 10 seeds, residual $\delta \in \{-1, 0, +1\}$, shaping $c = 0.5$). The **one change** is that the base rule is forced to `fixed`.
- **Code:** a new optional flag `env.base_rule_force` in a copy `n1/selectBaseRule_v17n.m`. When it is set, it skips the comparison. Default behaviour must stay bit-identical (test).
- **Training seeds:** fresh, base_seed 24042, blocks +201..+210.
- **Evaluation: M1's evaluation traces and CRN** (500001–500500 / 510001+). This deliberate reuse makes the paired comparison with M1's comparators and M1's oracle-base policy exact. It is a design ablation of the base rule, not a new claim about the regime.
- **Comparators:** stored M1 values for the 9 comparators (the calibrated tables are identical by construction; checked).

4. Pre-registered rules and wording

- **"Beats" machinery:** `common_rules.md` R1 (Holm over the 9 stored comparators) and the seed-level rule R2. Also a paired test of NR1 vs the M1 oracle-base primary on the 500 traces, and the NR1 vs M1 difference with its CI. If G6 has been run, the margins of NR1 against the G6 re-calibrated rules and the combined comparator are reported as a **secondary** analysis (G6 §3 family).
- **Definitions:** *NR1 WIN* = beats all 9 comparators. *NR1 Claim A only* = passes Claim A (OLLA family) but fails Claim B. *NR1 fails* = does not pass Claim A.

Outcome	Paper statement
NR1 WIN	"The gain does not require class-oracle information in the base rule: with a single global offset as base, residual A2C still beat every comparator (+x % vs the best rule)."
NR1 Claim A only	"Part of the margin over calibrated rules comes from the oracle base: with a non-oracle base the residual policy beat the OLLA family (+x %) but not the calibrated rules (margin y %, CI …)." Added to the abstract scope and to the limitations; C2 is qualified

Outcome	Paper statement
NR1 fails	A **stated limitation**: "With a non-oracle base the residual policy did not beat the OLLA family under realistic HARQ timing; the M1 result relies on the class-oracle base rule." C2 is qualified

5. Compute (basis 0.14 s per training episode, 0.095 s per evaluation)

- Training: 30,000 episodes ≈ **70 min**.
- Evaluation: (10 A2C + 10 best-on-val + 1 ensemble) × 500 ≈ **17 min**.
- Calibration and preflight ≈ 5 min.
- **Total ≈ 1.5 h.**

6. Multiplicity This adds 1 case (M1-NR) with up to 3 rows (`common_rules.md` R7). It is a design ablation.

6a. Stage specification for the runner (`Optimized/v17x/`, stage `nr1`) Implemented as `stage_nr1(ctx)`. Units: `calib` (must reproduce M1's calibrated tables; stop if not), `train_seed_K` (10 per-seed checkpoints, every 250 episodes), `eval`. Outputs `fill_summary_v17n.txt`, `table_nr1.csv`.

7. Claude Code prompt

> In `Optimized/v17/`, implement NR1 as in `paper/proposals/NR1_nonoracle_residual_prereg.md` (frozen). 1. Create `n1/selectBaseRule_v17n.m` (copy) with an optional `env.base_rule_force` ('fixed' | 'fixed_class'). Without the field, the output must be bit-identical to `selectBaseRule_v17` (write a test). 2. Add case `N1_dl_rtt8_fixedbase` to a copy `n1/v17n_setup.m`: M1's struct with `base_rule_force = 'fixed'`, base_seed 24042, and M1's `trace_seed_base` / `crn_seed_base`. Make `runComparison` use the new selector via the path order of a copied `n1/runComparison_v17n.m`. Change only the selector call. 3. Run a preflight (2 episodes, 2 seeds, 6 traces), then the full case. Verify that the calibrated `fixed_offset`, `class_tab` and `ctx_tab` equal M1's. Stop if not. 4. Compute REPLCMP/REPL/REPLSEED (Holm over 9) and a paired NR1 vs M1-primary comparison on the 500 traces. Write `Dataset/fill_summary_v17n.txt` and `README_v17n.md` with the verdict per §4. Branch `v17n`.

8. Results *(empty until run)*

G10_operating_grid_prereg.md

SHA-256: `ceecd80b8224a21d1665b45136413ad02871d4ae5cc7f356a6af671a9419131e`

Pre-registration G10: operating-point check at two neighbouring cells, with per-cell recalibration (v17c-grid)

STATUS: FROZEN (Checkpoint 5, 4 Nov 2026), amended at Checkpoint 5b (4 Nov 2026) and by Amendment 2 (5 Nov 2026), both before any G10 run started (`AMENDMENTS.md`: amendment 1: two cells instead of four, profile decided, wording rows for two cells; **amendment 2: the combined slack × class × HARQ-state comparator is not run in G10; it is tested at the M1 point only, in G6**). §§1–6 do not change after the freeze-v3 commit recorded in `paper/STATUS.md`; results go in §8. Common rules: `common_rules.md`. **Class: essential.** Uses the frozen v17 code; only case structs and calibration drivers are added.

1. Question The only earlier load/deadline grid (v14 Y1–Y4) predates the v17 PF fix and ran at RTT 0; its verdict was "operating-point specific". Does the M1 result (RTT 8, 8 processes, 12 kbit, 10-TTI budget) carry over to **two neighbouring operating points** under the v17 configuration, **against single-cue comparators (Fixed, FixedPU, Context table, tuned-target OLLA) that are re-calibrated for each cell**? The combined slack × class × HARQ-state comparator is **not** part of G10 (Amendment 2); its test belongs to G6 at the M1 point.

2. Cells All cells use M1's configuration (RTT 8, 8 processes, `pf_avg = 'scheduled'`, drain, useful bits) and residual A2C with 3000 episodes.

Cell	Packet bits	Deadline (TTI)	Status
G2_hi_d10	16,000	10	**tested**: higher load at M1's deadline (the tightest budget that lets a retransmission arrive at RTT 8)
G3_lo_d18	8,000	18	**tested**: lower load with a looser budget that allows two HARQ rounds
G1_lo_d10	8,000	10	**future work, decided now**
G4_hi_d18	16,000	18	**future work, decided now**

Why this pair (one line). G2 and G3 are opposite corners of the grid, so a win at both cannot be credited to a single axis and a loss at either locates the failing direction; G3_lo_d18 also sits where the older v14 grid (RTT 0, pre-v17 scheduler) found calibrated rules strongest (loose deadlines), which makes it the hardest test.

The other two cells are future work, decided now. They will **not** be added on the basis of what cells G2 and G3 show, and the paper lists them as untested. If a chosen cell is infeasible (below) it is **not replaced** by a future-work cell; the verdict is then computed on the remaining cell and the text says one cell was untestable.

Why 10 and 18, not v14's 6 and 14. With RTT 8 a 6-TTI budget lets no retransmission arrive in time. A budget of 18 (≤ the 20-TTI packet period) allows two feedback rounds.

Feasibility rule (v14 C2, unchanged): Fixed's on-time ratio on 20 calibration traces must lie in [0.50, 0.97]. Otherwise move the packet size one step towards 12,000 (8,000 → 10,000; 16,000 → 14,000). If the cell is still outside, it is infeasible and is not run.

- **RL:** 5 seeds per cell, 3000 episodes; base seeds 31042 (G2) and 32042 (G3), blocks +201..+205. **Primary scheme:** the 5-seed mean. The ensemble is secondary. Bandits (global, delay class) are trained on the same episodes as in v17.
- **Evaluation:** 300 traces per cell. Trace seeds 720001+ (G2) and 740001+ (G3); CRN at +10000. **Calibration traces:** 200 per cell, seeds 901001–901200 (G2) and 902001–902200 (G3); CRN 911001+ and 912001+ (the cell index k of the full four-cell design is kept, so adding a future cell would not change these ranges). The code asserts that all ranges are disjoint from v9–v17, from G6 (600001–600500) and from A6 (800001–800100).

2.1 Per-cell recalibration and the profile decision **Every single-cue comparator is re-calibrated under that cell's own load and deadline with the G6 method** (`G6_calibration_budget_prereg.md` §2.2, search on 200 calibration traces per cell): Fixed (exhaustive over the 9 offsets), FixedPU (separable grid plus joint refinement, at most 3 sweeps), Context table (exhaustive, 343 tables) and OLLA-tuned (30 settings). Calibration-set and evaluation-set values are reported side by side for every table. The search of every one of these four rules is **exhaustive or identical to G6**, so this section has no profile choice to make (the `lite` profile of G6 concerns only the combined table and is not used anywhere in G10).

Amendment 2: no combined comparator in G10. The combined comparator of G6 §2.3 (36 cells, coordinate ascent from four starts, re-run per cell) is **not calibrated or evaluated in any G10 cell**. Reason: re-calibrating it per cell dominates the compute of this stage (the corrected preflight projects about 100 h extra for two cells, `AMENDMENTS.md`), and a lighter variant of the search would give a weaker comparator than the one used in M1, so a win against it would be weaker evidence than the M1 result. The decision was taken before any G6 result existed and does not depend on one. **Consequence for what the paper may claim:** G10 can support statements about the OLLA family and the single-cue calibrated rules at the two neighbouring points. The statements "beats a rule that combines slack, class and HARQ-process state" and "same-information" are tested **at the M1 operating point only** (G6) and are never extended to other cells. Whether a combined rule closes the gap at the other cells is not tested and is listed as future work, decided now (it will not be added on the basis of what G2 and G3 show).

One budget difference from G6, stated plainly. Each G10 cell has **200** calibration traces, where G6 has 500. This is a compute-driven reduction of the calibration set size, applied to **all** comparators in the cell alike; the calibration objective is the mean on-time ratio over the 200 traces. The paper states that the G10 comparators were calibrated on 200 traces per cell, against 500 in G6 and 10–20 in v17.

Base rule for the residual: auto-selected as in v17 (the better of Fixed and FixedPU on 20 calibration traces). **Each cell records which base was chosen and whether it is a class oracle** (FixedPU) or not (Fixed), in `base_rule_by_cell.csv` and in the cell table of the paper.

3. Decision rules

- **Per cell:** family of **9 comparators** (the nine of v17, with Fixed, FixedPU, Context and OLLA re-calibrated for the cell), Holm over 9 (`common_rules.md` R1). Claim A and Claim B are reported per cell. **A cell is a WIN if the primary scheme beats all 9** (the v17 definition; there is no JP in G10). The fairness guard is applied to each cell.
- **Seed-level rule** (`common_rules.md` R2): per cell, the hierarchical seed × trace 95 % CI of the margin vs the best of the nine comparators. With 5 seeds it is indicative; a WIN whose lower bound is ≤ 0 is marked "not robust to seed resampling".
- **Cells tested** = the feasible cells of the pair that were run (`G10_CELLS`, default `G2_hi_d10,G3_lo_d18`). All wording below says how many cells were tested.

4. Pre-registered wording (two cells tested; the paper never says "robust across the load and deadline range")

Outcome	Paper statement
Both cells WIN (2 of 2)	"The gain held at both tested neighbouring operating points (a higher load at the M1 deadline, and a lower load with a looser deadline): residual A2C beat the OLLA family and the single-cue calibrated rules re-calibrated for each cell (margins +a % and +b % vs the best rule). The combined slack, class and HARQ-state rule was tested at the M1 point only. Two further grid corners were not tested."
One cell WINs (1 of 2)	"The gain is operating-point dependent: residual A2C beat the OLLA family and the single-cue rules at [cell] but not at [cell] (per-cell table). The combined rule was tested at the M1 point only. Two further grid corners were not tested." Also in the abstract's scope sentence

Outcome	Paper statement
No tested cell WINs	A **stated limitation**: “The result was observed at the M1 operating point only; at the two tested neighbouring points it was not reproduced (per-cell table). Two further grid corners were not tested.” Also in the abstract’s scope sentence
Only one cell feasible	The row above for 1 cell: “…at the one testable neighbouring point…”, and the text says the other chosen cell was infeasible under the feasibility rule
G10 not run	“The result is established at one operating point (12 kbit packets, 10-TTI budget); other loads and deadlines were not tested under the corrected scheduler.”

5. Compute (basis `compute_basis.md`; per cell; upper-bound rates; Amendment 2 removes the combined-comparator row)

Step (per cell)	Evaluations	Wall time (basis)
RL training: 5 seeds × 3000 episodes × 0.14 s	—	0.6 h
Evaluation: ≈ 8,400 runs (300 traces × 28 scheme runs: 7 deterministic, 5 A2C, 5 best-on-val, 1 ensemble, 10 bandit) × 0.095 s	8,400	0.2 h
Single-cue recalibration (Fixed + FixedPU grid 1,800; FixedPU joint ≤ 28,800; Context 68,600; OLLA 6,000)	≈ 105,200	2.8 h
Feasibility, counting, REPRO	≈ 2,000	0.1 h
Per cell		**≈ 3.7 h**
Two cells (this pre-registration)		**≈ 7.4 h**

(Before Amendment 2 the table also held the combined comparator, ≈ 9.1 h min / 13.7 h expected / 27.3 h cap per cell, giving ≈ 26 / 35 / 62 h for two cells.) The runner prints the projection from the measured rate (`PREFLIGHT_g10.txt`) and pauses if it exceeds 1.5× the basis (`common_rules.md` R5). `n_workers = 2`.

6. Multiplicity Adds **2 cases** (G2, G3) with 2 rows each (primary-scheme row and ensemble row). The check is a robustness check and cannot produce a new headline claim.

7. Stage specification for the runner (`Optimized/v17x/`, stage `g10`) Implemented as `stage_g10(ctx)`; per cell, in the order G2, G3: units `feas`, `calib_single_*`, `train_seed_K` (5, checkpoint every 250 episodes), `bandit_*`, `eval`. Outputs: `fill_summary_v17c.txt`, `base_rule_by_cell.csv`, `table_cells.csv` (calibration vs evaluation per table). `stage_g10_report(ctx)` writes a descriptive per-cell report (calibration information only) after each cell’s calibration. The stage contains **no** combined-comparator code path (Amendment 2). `G10_CELLS` defaults to the two tested cells; the future-work cells are not offered by the stage.

8. Results *(empty until run)*

A6_pf_timing_prereg.md

SHA-256: `35ffa58f563c5e184b5429acafcdbbcb4fa92f78d1ee4970c2609af4dca1a8d8`

Pre-registration A6: PF-timing demonstration for pitfall F8 (v17pf)

STATUS: FROZEN (Checkpoint 5b, 4 Nov 2026), before any run started (`AMENDMENTS.md`, amendment 1). Results go in §6 only. Common rules: `common_rules.md`. **Class: approved with the condition that it gets its own frozen text and joins the queue after G10.** It needs new evaluations (no training), so it is not a free analysis.

1. Question Does the timing of the PF average update (at transmission time vs at ACK time) change the goodput and fairness of the **same** schemes on the **same** traces, and does the effect depend on HARQ round-trip time and the number of processes? Today the ACK-time numbers come from a 6-trace scan (README_v17 §1) and the scheduling-time numbers from different 20-trace runs, so they cannot share one honest figure (ledger H17, H17a).

2. Design (descriptive; no claim about RL)

- **Frozen v17 code**, M3's full-buffer goodput environment, `drain = true`, `useful_bits = true` in **both** arms. The only factors: `pf_avg ∈ {'acked','scheduled'}` and HARQ ∈ {(RTT 0, 1 process), (RTT 8, 1 process), (RTT 8, 8 processes)}: 6 cells.
- **Schemes:** Baseline, OLLA (10 % target), Fixed (offset −1, M3's calibrated value). All three are deterministic.
- **Traces:** 100 traces, seeds **800001–800100**, CRN **810001+**, identical in all 6 cells (code asserts disjointness from v9–v17, G6, G10).
- **Reported per cell and scheme:** goodput, Jain index, edge5, with 95 % CIs over traces. Paired differences between the two PF modes at equal HARQ settings, with 95 % CIs.
- **Gate:** before the stage runs, `test_equivalence_v17.m` (B2.1: `acked` + `drain = false` reproduces v16 bit for bit) must pass in the same session.

3. Pre-registered expectation and wording (no decision rule on a hypothesis; the wording follows the measurement)

- **Expectation E:** under `acked`, 8 processes at RTT 8 give a goodput more than 10 % above 1 process at RTT 8 for each of the three schemes, and Jain falls by more than 0.02; under `scheduled`, the 8-process / 1-process goodput ratio at RTT 8 lies in [0.97, 1.03] and |ΔJain| ≤ 0.02.
- **If E holds for all three schemes:** F8 shows it, and the text says "the timing of the PF average, not HARQ itself, accounts for the apparent gain in this diagnostic (goodput +x % to +y % across schemes)".
- **If E holds for some schemes only:** the text reports the measured range per scheme and drops "every scheme".
- **If E does not hold:** F8 and the pitfall paragraph are revised to the measured values; the H17 6-trace figure is then quoted only as the original diagnostic.
- **If A6 is not run:** F8 becomes a row of T5 with H17 (+39.6 % Baseline, 6 traces) and H17a (10–26 % across schemes, confounded).

4. Compute (basis `compute_basis.md`) 3 schemes × 6 cells × 100 traces = 1,800 evaluations × 0.095 s ≈ **3 min**, plus the equivalence test (≈ 2 min) and the preflight.

5. Multiplicity and the registry A6 trains nothing and tests no hypothesis. It is added to the registry as **one case with role `diagnostic`** (no scheme-evaluation rows of an RL scheme): project total 73 cases after all runs (67 + W1 1 + G6 1 + NR1 1 + G10 2 + A6 1).

6. Stage specification for the runner (`Optimized/v17x/`, stage `a6`) `stage_a6(ctx)`, `stage_a6_preflight(ctx)`. Units: `equiv_test`, `cell_<pfmode>_rtt<R>_np<P>` (6). Outputs: `fill_summary_v17pf.txt`, `a6_pf_timing.csv` (per cell and scheme: goodput, Jain, edge5, CIs; paired differences). Gate: the equivalence test passes.

7. Results *(empty until run)*

common_rules.md

SHA-256: `f8382b866c5b51131d4185f4fe2920efabe46b6f7ed81963ec5759472fed664f`

Common rules for the Checkpoint-5 runs (G1, W1, G6, NR1, G10, A1, A3)

STATUS: FROZEN (Checkpoint 5, 4 Nov 2026), amended at Checkpoint 5b before any run started (`AMENDMENTS.md`), together with the pre-registrations that cite it (G6, W1, NR1, G10, A6).

R1. "Beats X" = two-sided Wilcoxon signed-rank on paired per-trace values, Holm-adjusted over the case's comparator family ($p < 0.05$), **and** the 90 % CI lower bound of the paired % margin > 0. Identical to `analyze_v17_replication.m`. Claim A, Claim B and WIN are defined as in `README_v17.md` §3.3, with the family size stated in each pre-registration.

R2. Seed-level rule. For every claim "beats the best comparator", the hierarchical seed × trace bootstrap (B = 20,000; seeds resampled, then traces) 95 % CI of the margin is reported, together with the sign test over the per-seed margins and the bootstrap over the seed margins (as in `paper/analysis/g3_seed_stats.py`). If a claim passes R1 but the hierarchical 95 % lower bound is ≤ 0, every statement of that claim in the paper adds "not robust to seed resampling" and the interval. For M1 vs Context and vs Fixed all seed-level lower bounds are > 0 (ledger G3a, G3g, G3h), so no such qualifier applies to M1 as it stands.

R3. No selection on evaluation traces. Calibration, base-rule choice, table search and every hyper-parameter use only calibration/validation traces. Calibrated tables are written to a file and hashed (SHA-256, logged) before any evaluation-trace run. A stage that needs the hash refuses to run without it.

R4. Determinism and pairing. Evaluation uses the traces and CRN draws given in each pre-registration. Frozen v17 evaluators are used unchanged, or an instrumented copy that passes a bit-identity test against the frozen one before use. Training seeds and trace ranges are disjoint from all earlier rounds (the code asserts it).

R5. Preflight and pause rule. Every stage runs a 20-trace preflight and measures seconds per unit (one 200-TTI trace evaluation, or one training episode). Basis (`compute_basis.md`): 0.095 s per evaluation, 0.14 s per training episode, 2 workers, RTT 8. The runner prints the projected stage time. If the measured rate exceeds **1.5×** the basis, the stage **pauses**: it writes `PAUSED_<stage>.txt` and exits without starting. The author resumes by creating `APPROVE_<stage>.txt`. A measured rate below the basis never pauses and only scales the projection down.

R6. Failure handling. A stage that errors writes `ERROR_<stage>.txt` (message and stack) and **no** "done" marker. It is retried at most once after a restart; if the second attempt also errors, the pipeline halts (default) and the status file reads `HALTED`. A native crash (no marker written) triggers a restart of the whole runner from the checkpoints. "FINISHED" is written only when every requested stage has a done marker.

R7. Multiplicity bookkeeping. Each stage's new registry rows are listed in its pre-registration: W1 1 case, G6 1, NR1 1, G10 2, A6 1 (diagnostic), giving 73 cases in all; the scheme-evaluation rows rise from 108 to at most 119 (W1 2, G6 2, NR1 3, G10 4) plus the A6 diagnostic row. All are robustness checks, design ablations or diagnostics of M1, never new claims. M1's Bonferroni factor is at most 1,190 tests (119 rows × 10 comparators); its p of 9.06e-13 gives ≈ 1.1e-9.

R8. Parallelism. 2 parfor workers (`n_workers = 2`). A larger pool only by explicit author decision (4 workers ran out of memory in v17 M1).

R9. Report points. After every stage the runner calls the optional `stage_<id>_report(ctx)` and writes `Dataset\REPORT_<id>.txt`; the pipeline **continues without waiting**. A report is descriptive: it may contain only information the stage was already pre-registered to produce, and it **cannot change any pre-registered rule, design choice or wording**. A failing report function is logged as a warning and does not invalidate the stage. Reports of calibration stages contain calibration-set information only (R3).

checkpoint5_list.md

SHA-256: `9debb4c685cf84a2af6afd787a6c84b3f95f061ccf8f0e58b80321b3b002909c`

Checkpoint 5: consolidated evidence-gap list (Phase 5)

Compute basis: `compute_basis.md` (RTT 8, 2 workers: about 0.14 s per training episode and 0.095 s per 200-TTI evaluation; upper bounds). Every run has a 20-trace preflight and pauses if the measured rate exceeds 1.5× the basis. Each pre-registration's §§1–6 are frozen on approval.

Summary (FROZEN; amended at Checkpoint 5b, see `AMENDMENTS.md`) All pre-registrations G6, W1, NR1, G10, A6 and `common_rules.md` are frozen; the freeze commits and hashes are in `paper/STATUS.md` and `FREEZE.sha256`. **Run order (approved at Checkpoint 5b): G1 → W1 → NR1 → G6 (g6a → report point → g6b → g6c) → G10 → A1/A3 → A6.** A2, A4 and A5 need no runs. NR1 runs before G6 so that the short items report early.

ID	Item	Class	New runs?	Wall time (upper-bound basis; 2 workers)	Pre-registration
G1	Action re-cut on scheduled new transmissions (M1, M3)	essential	re-evaluation	20–60 min	`paper/analysis/G1_SPEC.md`
G3	Seed-level statistics	essential	none	—	**done** (ledger G3a–G3i)
W1	Warm-up robustness (200 unscored TTIs, stored policies)	essential	re-evaluation	≈ 1.2 h	`W1_warmup_prereg.md`
NR1	Residual on the non-oracle Fixed base at RTT 8	recommended	training	≈ 1.5 h	`NR1_nonoracle_residual_prereg.md`
G6	**Full profile.** Exhaustive re-calibration of Fixed / Context / OLLA on 500 traces, FixedPU separable + joint, and the combined comparator (coordinate ascent, 4 starts)	essential	calibration, no RL training	`g6a` ≈ 6.9 h (**report point**), `g6b` ≈ 23 h min / **≈ 34 h expected** / ≤ 68 h cap; **total ≈ 41 h expected**	`G6_calibration_budget_prereg.md`

ID	Item	Class	New runs?	Wall time (upper-bound basis; 2 workers)	Pre-registration
G10	**Two cells** (G2_hi_d10, G3_lo_d18), per-cell recalibration of the **single-cue** comparators (Fixed, FixedPU, Context, OLLA-tuned), 200 calibration traces per cell; **no combined comparator** (Amendment 2; tested at M1 in G6 only)	essential	training + calibration	**≈ 7.4 h on the basis** (≈ 3.7 h per cell; before Amendment 2: ≈ 35 h expected)	`G10_operating_grid_prereg.md`
A1	Results by speed class (M1)	approved	re-evaluation	≈ 20 min	below
A2	On-time attribution by state (from the G1 log)	approved	none	minutes	after G1
A3	Inference wall time per TTI	approved	micro-benchmark	≈ 5 min	below
A4	Learning curves and entropy (HDF5)	approved	none	—	Phase 7 figure F5b
A5	Path-agnostic release setup	approved	none	—	below
A6	PF-timing demonstration (pitfall figure F8)	approved **with its own frozen text, after G10**	re-evaluation (no training)	≈ 3 min + 2 min test	`A6_pf_timing_prereg.md`
F1, F2	SALAD comparator; second simulator	future work	—	—	—
F3, F4	Grid corners G1_lo_d10 and G4_hi_d18	future work, decided now (not to be added on the basis of G2/G3)	—	—	`G10_operating_grid_prereg.md` §2

Total on the upper-bound basis: ≈ 1 h (G1) + 1.2 h (W1) + 1.5 h (NR1) + 41 h (G6) + 7.4 h (G10, Amendment 2) + 0.5 h (A1/A3) + 0.1 h (A6) ≈ **53 h on the basis** (before Amendment 2: ≈ 80 h). The measured preflight rates for the calibration steps are higher than the basis (author-reported, 5 Nov 2026); `PREFLIGHT_<stage>.txt` holds the measured projections. The measured preflight rate (expected below the basis for deterministic rules) can only shorten it.

Multiplicity after all runs: registry +6 cases (W1 1, G6 1, NR1 1, G10 2, A6 1 diagnostic) → **73 cases**; none is a new claim. M1's Bonferroni factor is at most 1,190 tests (`common_rules.md` R7); M1's p (9.06e-13) gives ≈ 1.1e-9.

TWC revisit rule (Checkpoint 3): only if G6 (A, B-500 and JP all pass, seed-level bounds > 0) **and**

G10 (both tested cells WIN) are strongly positive.

Seed-level rule (applies everywhere): `common_rules.md` R2. For M1 all seed-level 95 % lower bounds vs Context are > 0 (+0.72, +0.48, +0.62; ledger G3g).

Prompts for the items without their own file G1 (already given): > In `Optimized\v17`, run `addpath('g1'); g1_recut_v17` (default cases M1, M3). Report `paper/g1/g1_summary.txt` and `g1_action_by_state.csv` back. If `EQUIV` or `REPRO` fails, stop and report. Do not edit `evalPolicy_v17.m`.

A1: results by mobility regime (pre-registration, short): - *Question:* how does M1's margin vs the Context table and vs Fixed vary with the user's speed class (< 10, 10–20, > 20 m/s, i.e. f_d ≈ 20–67, 67–133, 133–200 Hz)? - *Design:* re-evaluate the stored M1 policies (10 seeds), Context, Fixed and FixedPU on the 500 M1 traces. Record per-user on-time counts (`info.okU`, `info.resU`) and per-user mean speed. Aggregate on-time per speed class. Use the hierarchical bootstrap CI per class. - *Status:* descriptive only. No new claim; it backs the "report by mobility regime" recommendation (C4). > Prompt: "In `Optimized/v17`, write `a1/a1_by_speed.m`: regenerate M1's evaluation traces exactly as `runComparison_v17.m` L183–191 does; evaluate `Deep A2C v17` (10 stored seeds), `ctxtab`, `fixed` and `fixed_class` with the frozen `evalPolicy_v17.m`; save per trace and per user `okU`, `resU`, mean speed and true delay to `paper/a1/a1_peruser.mat`; REPRO-check cell goodput against the stored `metrics.cell_tp`. No statistics in MATLAB; the Python bootstrap is done on the cloud side."

A3: inference wall time: > "In `Optimized/v17`, write `a3/a3_timing.m`: load seed 1 of the M1 policy, extract the actor weights as `diag_action_by_state_v17.m` does, and time 10^5 forward passes for N = 10 users (input 57) with `timeit`. Also time one full `evalPolicy_v17('a2c_v17', …)` 200-TTI trace vs `'fixed_class'`. Report the median µs per decision and per TTI, plus the CPU model."

A5: path-agnostic release setup: > "Create `Optimized/v17/release/code/v17_setup_portable.m` that derives `root` from `fileparts(mfilename('fullpath'))` instead of the hard-coded `D:\` path, and document it in `README_release.md`. Do not change `v17_setup.m` (frozen); add a test that both give identical `C` and `base` structs."

Changes to the paper plan implied by these items

- C2's wording depends on G6 (rule in G6 §4) and, if run, on NR1 (§4 there).
- The abstract's gain-vs-OLLA number depends on W1 (§3 there).
- The behaviour paragraph (Results VI-G) depends on G1, and optionally A2.
- The robustness paragraph depends on G10. If G10 is not run, the paper states the operating-point restriction explicitly.

compute_basis.md

SHA-256: `d72e7a0f64506fe278ed319bc0bbe74d564de0cf1a27c3543311acd226fade0a`

Compute basis for all Checkpoint-5 estimates

Measured on the author's PC: MATLAB R2023a, 16 GB RAM, `n_workers = 2`. Times are the `STEP|…|done|elapsed_min` lines in `Optimized/v17/Dataset/fill_summary_v17.txt`.

Case	Wall time (min)	Training episodes	200-TTI trace evaluations
M1_dl_rtt8	160.8 (includes one interrupted launch)	10×3000 + 10×800 + 10×800 = 46,000	(7 det. + 10 A2C + 10 best-on-val + 1 ens. + 10 offset + 10 resid-800 + 20 bandit) × 500 = 34,000
M3_gp_rtt8	70.0	10×800 + 5×800 = 12,000	(7 + 10 + 10 + 1 + 5 + 20) × 500 = 26,500

Solving the two linear equations gives, for RTT 8 on 2 workers: - **≈ 0.14 s wall per training episode** (200 TTI, including A2C updates and bandit training); - **≈ 0.095 s wall per 200-TTI trace evaluation** (any scheme, including calibration runs).

These are **upper-bound** rates, because M1 includes an interrupted launch. A 400-TTI trace (200 warm-up + 200) counts as about 2 evaluations. RTT-0 cases run faster (M4: 56 min for fewer evaluations). Every proposal includes a 20-trace preflight that measures the true rate before the main run. If the preflight rate is more than 1.5× the basis, the run pauses for the author.

NR2_confirmatory_prereg.md

SHA-256: 1210d5b17aeae883796ad369f17e7ef1ffeed404d7c317c23d7b4d97612c3e0c

Pre-registration: NR2, confirmatory replication of NR1 on fresh data (v17x stage nr2)

STATUS: FROZEN (8 Nov 2026, Amendment 3). §§1–8 and the analysis script paper/analysis/nr2_stats.py do not change after the freeze commit recorded in paper/STATUS.md and FREEZE.sha256. Results go in §10. Common rules: common_rules.md (R1–R9 apply unless stated otherwise here). **Run order: after G10.**

Independence. Written after NR1, G6, W1, A1, A3, A6 and G1 results were known (they motivate it), and before any NR2 run, preflight or trace generation. No NR2 trace, seed or output exists at freezing.

1. Why NR2

- NR1 (non-oracle Fixed base, offset −2) beat all 9 comparators and was +1.77 % above the M1 oracle-base policy (table_nr1.csv). A descriptive comparison put it +2.44 % [2.04, 2.85] above the G6 combined table (paper/analysis/out/nr1_g6_compare.csv).
- Three weaknesses stop NR1 from carrying a recommendation: (i) it was evaluated on **M1's evaluation traces**, on which the G6 comparison was also run; (ii) the NR1-vs-combined-table comparison was **secondary and descriptive**; (iii) stage_nr1.m saved only the seed mean per trace, so **Rule R2 cannot be applied**.
- M1 vs the combined table passes at trace level but **fails R2** (hierarchical 95 % CI [−0.30, +1.70] %; ledger G6b/G6e). Whether a configuration exists whose margin over the strongest calibrated rule is seed-robust is the open question of the paper.

2. Question On fresh evaluation traces and fresh training seeds, does residual A2C on the **non-oracle Fixed base** beat the **frozen G6 combined table (Context-joint-500)** at trace level (R1) **and** at seed level (R2)?

3. Design (one change from NR1: fresh seeds and fresh evaluation traces)

- **Configuration:** identical to NR1 (n1/ code path, env.base_rule_force = 'fixed'), i.e. M1's environment: RTT 8, 8 HARQ processes, pf_avg = 'scheduled', drain, useful bits, deadline traffic (12 kbit / 20 TTI, budget 10 TTI), state tier O, residual $\delta \in \{-1, 0, +1\}$, shaping c = 0.5, deviation penalty 0.02, logit bias 3, hyper-parameters of v17_setup.m.
- **Base rule:** Fixed with offset **−2, frozen**. It is not re-calibrated. The stage asserts that −2 equals both the NR1 calibrated offset (ckpt/nr1/calib.mat) and the Fixed-500 offset (cal500_single.txt); it stops otherwise.
- **Training: 3000 episodes × 10 fresh seeds.** base_seed = 25042, seed blocks +201..+210 (seeds 25243–25252), the NR1 convention. Validation traces as in M1/NR1 (the training-time validation split, never evaluation traces).
- **Evaluation traces: 500 fresh traces**, rng(1000000 + j) for the channel, rng(1010000 + j) for the CRN matrix U, rng(1020000 + 100 j + 10 k + s) for scheme k, seed s, j = 1…500.
- **Disjointness (asserted in code, R4):** the trace, CRN and decision ranges above must not intersect any range used so far: v9–v17 cases (evaluation 500001–570500 and CRN; calibration from 60001; validation from 80001; decision draws 77000+), G6 calibration 600001–600500 / 610001+, G10 720000–750500 and 901000–912500, A6 800001–810100, and the training seed blocks 20042–24252 and 31042–32252. The stage stops if any intersection is found.
- **Comparators (frozen, never re-searched):** loaded with the loaders of stage_g6c.m. cal500_tables.txt must have SHA-256 94140b8c3405dc7107c2ef1b3d833fd2811fde47b2046516b8547ca331257f89 and cal500_single.txt 8ec52905dce4a61dcc9da329f9bd113068147fa0472ae5941ad0c43ec5821998; the stage refuses to run otherwise.
- **REPRO gates before any fresh trace is generated:** (a) the eight comparators re-evaluated on M1 evaluation traces 1–20 reproduce ckpt/g6c/eval.mat (G6 rules) and results_v17_M1_dl_rtt8.mat (Baseline, OLLA, Multi-OLLA) **bit-identically**; (b) the stored M1 policies (10 seeds) re-evaluated on traces 1–20 reproduce the stored M1 per-trace values bit-identically. Any difference stops the stage.

4. Comparator family and decision rules

- **Primary scheme:** the **per-seed mean**: each of the 10 NR2 policies is evaluated on all 500 traces, and the per-trace on-time ratio is averaged over seeds.
- **Family (Holm over 8):** Baseline, OLLA, OLLA-tuned-500, Multi-OLLA, Fixed-500 (= the NR2 base rule), FixedPU-500, Context-500, **Context-joint-500 (primary comparator)**. The bandits are not in the family (they would need training on NR2's episodes; they were never close).
- **Trace level (R1):** margin = 100 · mean(d) / mean(b) with d the paired per-trace difference; 90 % t-interval of mean(d) scaled by mean(b); two-sided Wilcoxon signed-rank; Holm over the 8. "Beats X" = Holm $p < 0.05$ **and** 90 % CI lower bound > 0.
- **Seed level (R2):** vs Context-joint-500, the hierarchical bootstrap (B = 20,000; resample seeds, then traces with CRN pairing kept; RNG seed 20261104), the bootstrap over the 10 seed means (RNG seed 20261105) and the one-sided sign test over seed margins, all as in `paper/analysis/g3_seed_stats.py`. **Seed-robust** = hierarchical 95 % lower bound > 0.
- **Verdict (vs Context-joint-500):**
  - **CONFIRM** = beats at trace level **and** seed-robust.
  - **TRACE** = beats at trace level, not seed-robust.
  - **REVERSE** = Holm $p < 0.05$ and the 90 % CI upper bound < 0.
  - **NULL** = none of the above.
- **WIN** (reported with every verdict) = beats all 8.
- **Secondary (reported, not part of the verdict):** NR2 vs the M1 oracle-base policies re-evaluated on the fresh traces (paired, R1 machinery, no Holm), with the hierarchical CI; the deterministic ensemble of the 10 NR2 seeds; A1-style speed-class margins vs Context-joint-500 (hierarchical 95 % CIs).
- **Reframing gate** (`NR2_reframing_proposal.md`): CONFIRM **and** WIN.
- All statistics are computed by the frozen script `paper/analysis/nr2_stats.py` from `Optimized/v17x/Dataset/ckpt/nr2/eval.mat`; the MATLAB stage writes the data and its own trace-level table, and the two must agree: margins and CIs to two decimals, p-values within 5 % relative (else the stage is reported as inconsistent and no wording is selected).
- **Script validated before freezing:** run on the M1 per-seed values and the G6 comparator values, `nr2_stats.py` reproduces `table_cal500.csv` (margins and CIs exactly; Holm p within 1 % relative), the G6 seed-level result (5/10 seeds, hierarchical 95 % CI [−0.30, +1.70] %) and returns verdict **TRACE** for M1, as the corrected G6 reading requires.

5. Pre-registered wording (`paper/manuscript/wording.tex`, slot NR2)

Verdict	Paper statement	Further effect
CONFIRM	"On fresh evaluation traces and fresh training seeds, the residual policy on a non-oracle base beat the frozen combined table at trace level and robustly across training seeds (+x %, hierarchical 95 % CI [a, b] %)."	If WIN as well: the reframing proposal may be put to the authors (§9). Otherwise C2 gains this sentence only
TRACE	"… beat the frozen combined table at trace level (+x %), but not robustly across training seeds (hierarchical 95 % CI [a, b] %)."	No reframing. NR1 and NR2 are reported as consistent at trace level; the R2 qualifier stays in the abstract
NULL	"… the margin … over the frozen combined table was not significant (+x %, 90 % CI [a, b] %); the NR1 margin did not replicate."	NR1's secondary comparison with the combined table is described as not replicated; limitation added
REVERSE	"… the frozen combined table beat the residual policy on a non-oracle base (x %); the NR1 margin did not replicate."	As NULL, and the abstract states that a calibrated rule beat the non-oracle residual on fresh data
not run	"NR2 had not been run at the time of writing."	—

Numbers are filled from `nr2_stats.py` output into `pending_values.csv` (item NR2; keys in `pending_manifest.csv`). No other wording may be used for the NR2 result.

6. Compute (measured basis, not the 0.14 s/episode basis)

- Basis: the NR1 run on the same PC. Training 1044–2123 s per seed (median ≈ 1450 s); NR1 preflight 0.64 s per episode. Evaluation ≈ 0.07 s per 200-TTI trace (NR1: 731 s for 21 × 500 evaluations).
- Training: 10 × 3000 episodes ≈ **4.0 h** (range 2.9–5.9 h).
- Evaluation: (10 NR2 + 1 ensemble + 10 M1 + 8 comparators) × 500 = 14,500 evaluations ≈ **0.3 h**; REPRO gates ≈ 0.05 h; statistics (python) ≈ 0.05 h.
- **Total ≈ 4.4 h (range 3.3–6.3 h).** R5: the preflight basis for this stage is **0.64 s per episode**; the stage pauses if the measured rate exceeds 0.96 s.

7. Multiplicity One new case (M1-NR2) with up to 3 registry rows (NR2 primary, NR2 ensemble, M1 re-evaluated). It is a confirmatory replication of a design ablation. The registry rows of the v17x stages, including NR2, are added before the counts in the paper are final.

8. Stage specification for the runner (`Optimized/v17x/`, stage `nr2`) `stage_nr2(ctx)`, with `stage_nr2_preflight` (20 training episodes on seed 1 and 20 evaluation traces drawn from a **separate** preflight range 1030001–1030020, never from the evaluation range) and `stage_nr2_report` (R9: descriptive, calibration/REPRO information only). Units (each checkpointed, R6): 1. `check` — SHA-256 of both table files; base offset −2 equal in `ckpt/nr1/calib.mat` and `cal500_single.txt`; disjointness assertions of §3; `n1/selectBaseRule_v17n` default behaviour bit-identical (existing test). 2. `repro` — the REPRO gates of §3 on M1 traces 1–20. 3. `train_seed_K`, K = 1...10 — `DeepA2C_Shared_v17` with `base_seed + 200 + K`, 3000 episodes, checkpoint every 250 episodes; save the policy and learning curve. 4. `eval` — generate the 500 fresh traces; evaluate NR2 seeds 1–10, the NR2 ensemble, the 10 stored M1 policies and the 8 comparators. **Save per-seed per-trace arrays.** 5. `analysis` — trace-level table (`run_cmp_stats` of `stage_g6c.m`, Holm over 8), then call `python3 paper/analysis/nr2_stats.py` (or write `NEEDS_PYTHON_nr2.txt` and finish if python is unavailable; the statistics are then run by hand with the same script).

Outputs (`Optimized/v17x/Dataset/`): `ckpt/nr2/eval.mat` with `out.nr2_seeds` (10×500), `out.nr2_ens` (1×500), `out.m1_seeds` (10×500) and `out.cmp.<name>` (1×500) for `baseline, olla, olla_tuned500, multi_olla, fixed500, fixedpu500, ctx500, ctxj500`; per-user arrays `okU/resU/speeds` as in `stage_a1.m` (for the secondary speed breakdown); `table_nr2.csv`; `fill_summary_v17nr2.txt` (CONFIG, PRIMARY, REPLCMP lines, REPRO lines, hashes); `stage_nr2.done`. The script writes `paper/analysis/out/nr2_stats.csv`, `nr2_seed_stats.csv`, `nr2_verdict.txt` and `nr2_pending_values.csv`.

9. Claude Code prompt (for the runner on the PC)

> In `Optimized/v17x/`, implement stage `nr2` exactly as in `paper/proposals/NR2_confirmatory_prereg.md` (frozen; verify with `cd paper/proposals && sha256sum -c FREEZE.sha256`). Copy `stage_nr1.m` to `stage_nr2.m` and change only: seeds (`base_seed = 25042`), the fresh trace/CRN/decision ranges of §3 with the disjointness assertions, the frozen base offset −2 (no calibration unit; assert instead), the `check` and `repro` units, the 8-comparator family loaded with the loaders of `stage_g6c.m` (hash-checked), evaluation of the stored M1 policies, and the outputs of §8 (save per-seed arrays). Add `stage_nr2_preflight.m` (preflight range 1030001+, basis 0.64 s/episode) and `stage_nr2_report.m`. Register `nr2` in `run_v17x_all.m` **after `g10`**. Run the existing self-tests and a smoke run (`ctx.smoke`: 2 seeds, 2 episodes, 4 traces from the preflight range) and show me the outputs before any full run. Do not edit anything under `paper/`.

10. Results *(empty until run)*

AMENDMENTS.md

SHA-256: `1b370c55fb899ef2769bbdbe7f5436b709a5ca31b3b501fbf67283e375fc7017`

Amendments to the frozen pre-registrations

The freeze rule (`common_rules.md`, STATUS.md) lets a text change only **before any run that uses it has started**, and every change is recorded here with its reason. No stage of the Checkpoint-5 programme has been run: the runner stages are stubs (`Optimized/v17x/`).

Amendment 1: Checkpoint 5b, 4 Nov 2026 (author's decisions) Freeze v1: commit `0726555fb7dda0190e780406828f785903d98f30`. Freeze v2: commit recorded in `paper/STATUS.md` and `paper/proposals/FREEZE.sha256`.

File	Change	Reason
`G6_calibration_budget_prereg.md`	(a) the composite table is defined in full in §2.3 (formula, indices, when it is computed, its three uses); (b) profile fixed to **full**, with the exact differences of `lite` stated in §2.4; (c) new §2.5 report point after `g6a`	the sparse-cell fallback must be defined in the frozen text, not only in code; C2's wording depends on G6, so it gets the strongest search; the author wants early, descriptive information
`G10_operating_grid_prereg.md`	four cells → **two tested cells** (G2_hi_d10, G3_lo_d18); G1_lo_d10 and G4_hi_d18 are future work, decided now; profile decision and the 200-vs-500 calibration-trace difference stated; wording rows for two cells; compute for two cells; multiplicity 2 cases	the pair spans both axes in opposite directions and includes the regime where calibrated rules were strongest; programme length (≈ 3 days)
`A6_pf_timing_prereg.md`	**new frozen text**	A6 needs new evaluations, so it joins the queue after G10 with its own pre-registration
`common_rules.md`	R9 (report points); R7 case counts	descriptive reports after each stage; new totals
`checkpoint5_list.md`	run order G1 → W1 → NR1 → G6 → G10 → A1/A3 → A6; G10 two cells; totals	author's order and profile decisions
`W1`, `NR1`, `compute_basis.md`	unchanged	—

Run order after the amendment: **G1 → W1 → NR1 → G6 (g6a → report point → g6b → g6c) → G10 → A1/A3 → A6.** Expected programme length on the upper-bound basis: ≈ 80 h (about 3.3 days).

Amendment 2: 5 Nov 2026 (G10 compute; before any G6 result and before any G10 run) **Trigger.** The corrected preflight, reported by the author, shows that G10 with two cells costs about **100 h extra** because the combined comparator is re-calibrated per cell (36 cells, four starts, coordinate ascent to convergence). The basis in `G10` §5 had this step at 9.1–27.3 h per cell. Only the combined comparator is affected by the choice below; training, evaluation and the single-cue calibration are not.

Independence from results. The choice uses compute and claim scope only. No G6 output (`REPORT_g6a.txt`, tables, evaluation) was read by the writer of this amendment. `G6_calibration_budget_prereg.md`, `W1`, `NR1`, `A6` and `common_rules.md` are **unchanged**, so the running G6 stages are not touched.

Options considered. Hours are on the preflight scale implied by the author's figure, which the writer read as ≈ 135 h for two cells at the old design (basis: ≈ 35 h expected, factor ≈ 3.9 applied uniformly; replace by the numbers in `PREFLIGHT_g10.txt`; the ratios between options do not depend on the factor).

Option	What G10 does	Compute (basis, two cells → scaled)	What the paper can claim	Main weakness
(a) `lite` combined table in G10 cells, full exhaustive search for single-cue rules	per cell: 4 starts to convergence on a small subset (20 % of the 200 calibration traces = 40), best two polished on all 200	≈ 9.9 h per cell → ≈ 20 h → **≈ 77 h**	per cell: beats the OLLA family, the single-cue rules, **and a combined rule calibrated with a lighter search than in M1**	saves little; the strongest comparator is weaker than in M1, so a win is weaker evidence, and the 5b ruling excluded profiles that lower the calibration traces; sparse threshold must be rescaled for 40 traces
(b) single-cue rules only in G10; combined comparator at M1 only (G6)	per cell: Fixed, FixedPU, Context, OLLA re-calibrated (full, 200 traces); no combined-comparator code	≈ 3.7 h per cell → ≈ 7.4 h → **≈ 29 h**	two points: beats the OLLA family and the single-cue calibrated rules, or not; “same-information” and “beats the combined rule” are claims about M1 only	the combined rule might close the gap at the other cells; this is not tested (limitation, future work)
(c) one cell (G3_lo_d18) at full profile	all comparators incl. combined, full G6 method	≈ 17.4 h → **≈ 68 h**	one neighbouring point, against every comparator incl. the combined one; no statement about which axis matters	one point cannot separate load from deadline; programme still long

Decision recommended and applied: (b). It keeps both neighbouring cells and both axes, which is what lets a loss locate a direction (the reason the pair was chosen at 5b), uses the same exhaustive search as G6 for every rule it does include, and costs about a quarter of the old G10. The combined comparator is where the headline wording of C2 is decided; that decision is made at M1 (G6), where the search is full and the calibration set is 500 traces, and not extended to other points. (a) saves too little and weakens the strongest comparator; (c) keeps the strongest comparator but loses the second cell and still costs about half of the old G10.

File	Change	Reason
`G10_operating_grid_prereg.md`	§1 question names the single-cue comparators; §2.1 rewritten (no combined comparator, no profile choice left, consequence for claims, future work decided now); §3 family of 9, Holm over 9, WIN = beats all 9, no JP; §4 four wording rows carry “single-cue rules” and “combined rule tested at M1 only”; §5 compute (≈ 3.7 h per cell, 7.4 h for two); §7 units and outputs without the combined comparator	compute
`checkpoint5_list.md`	G10 row and totals (≈ 53 h on the basis; was ≈ 80 h)	consistency
`Optimized/v17x/README_v17x_runner.md`	g10 units without `calib_comb_*`	runner spec

File	Change	Reason
paper/manuscript/wording.tex, paper/design/outline.md	G10 wording variants and outline row follow §4	consistency
G6, W1, NR1, A6, common_rules.md, compute_basis.md	unchanged	G6 is running

Freeze v3: hashes in FREEZE.sha256; the commit is recorded in paper/STATUS.md. If the author prefers (a) or (c), the text can be replaced before any G10 run starts, with a further amendment entry (G10 has not started; the runner stage is a stub).

Amendment 3: 8 Nov 2026 (new stage NR2; no frozen text changed) Trigger. The review of the Checkpoint-5 results (paper/ledger/conflicts.md §G) found that (i) M1's margin over the G6 combined table passes at trace level but fails Rule R2 (5/10 seeds; hierarchical 95 % CI [−0.30, +1.70] %), and (ii) NR1 (non-oracle Fixed base) beat every comparator, but on M1's evaluation traces, with only a descriptive comparison against the combined table, and without stored per-seed values, so R2 cannot be applied to it. The author asked for a confirmatory replication of NR1.

What is added. NR2_confirmatory_prereg.md (new, frozen) and its analysis script paper/analysis/nr2_stats.py (frozen with it). Fixed base, the residual recipe, 3000 episodes, 10 fresh seeds (base 25042), 500 fresh evaluation traces (1000001–1000500, CRN 1010001+, decision draws 1020000+; disjoint from every earlier range, asserted in code), primary comparator **Context-joint-500 from the frozen G6 tables (hash-checked, not re-searched)**, primary scheme the per-seed mean, decision at trace level (Holm over a family of 8) **and** seed level (R2). Verdicts CONFIRM / TRACE / NULL / REVERSE with a wording row each (wording.tex, slot NR2); WIN reported with each. Compute ≈ 4.4 h on the measured NR1 basis (range 3.3–6.3 h). **Runs after G10.**

What is not changed. No frozen text changes: G6, W1, NR1, G10, A6, common_rules.md, checkpoint5_list.md and compute_basis.md keep their v3 hashes. NR2 is listed here and in FREEZE.sha256 rather than by editing checkpoint5_list.md. R7 bookkeeping: +1 case (M1-NR2), up to 3 rows.

Independence. NR2 was designed after the NR1 and G6 results were read; that is why it uses fresh traces and seeds and fixes the comparator, the scheme, the family and both decision levels in advance. The script was validated before freezing on the M1/G6 data, where it reproduces table_cal500.csv and returns TRACE.

Related decision (not a freeze change). The G6 stage's summary file marked the claim seed-robust because it evaluated R2 against Context-500 only; G6 §3 and R2 require the best comparator. The manuscript uses the G6 §4 row 2 wording (ABJ-seed); the generated file is not edited (ledger G6e).

Freeze v4: hashes in FREEZE.sha256 (v3 hashes kept in FREEZE_v3.sha256); the commit is recorded in paper/STATUS.md.

Amendment 4: 9 Nov 2026 (Code review finding: fixed_class_tab bug in comparator evaluation and calibration; watchdog heartbeat in G10; before any valid G10 or NR2 run) Trigger. A systematic code review of the live v17/v17x pipeline (claude/amc_code_review_2026-11-08.md) identified that evalPolicy_v17_g6.m evaluated policy 'fixed_class' exclusively by reading env.fixed_class_tab, whereas the calibration and evaluation stages (stage_g6a.m, stage_g6c.m, stage_g10.m, and stage_nr2.m) wrote the calibrated table to env.class_tab. Consequently, the G6c evaluation evaluated M1's historical 20-trace table ([-2 -2 -2; -3 -2 -3]) under the label FixedPU-500 instead of the calibrated 500-trace table ([-1 -2 -2; -1 -2 -2]). In G10, the same mismatch prevented per-cell FixedPU refinement and corrupted FixedPU base-rule selection. In addition, G10 was repeatedly terminated by watchdog_v17x.ps1 (14 times) due to a 30-minute watchdog timeout during the exhaustive 343-candidate Context search because heartbeats were not emitted during long optimization loops.

What is fixed. 1. evalPolicy_v17_g6.m: updated so that policies 'fixed_class', 'fixed_oracle', 'fixed_online', and residual base selection inspect env.class_tab first, falling back to env.fixed_class_tab. 2. stage_g6a.m, stage_g6c.m, stage_g10.m, and stage_nr2.m: explicitly assign both env.fixed_class_tab and env.class_tab across all calibration and evaluation units. 3. stage_g10.m: added throttled heartbeat calls (ctx.hb(...)) within Context candidate search (every 10 candidates), OLLA search (every 5 candidates), and coordinate ascent sweeps (every cell). 4. stage_g6c.m: re-evaluated on the 500 held-out traces with the corrected FixedPU-500 table. The true FixedPU-500 margin is +1.73 % [1.31, 2.14] % (Holm p = 2.79e-12; hierarchical 95 % CI [+0.72, +2.80] %, 9/10 seeds > 0). Residual A2C beats all 10 comparators. Calibrated rule order is strictly monotonic: Fixed-500 (+1.87 %) > FixedPU-500 (+1.73 %) > Context-500 (+1.43 %) > Context-joint-500 (+0.66 %). 5. stage_nr2.m: inherits the fix so that the REPRO gate and fresh evaluations correctly evaluate the true calibrated FixedPU-500 table. 6. G10 checkpoints were reset so G10 begins cleanly from the start before any valid results are produced.

What is not changed. The pre-registration specifications (G6_calibration_budget_prereg.md, G10_operating_grid_prereg.md, NR2_confirmatory_prereg.md, common_rules.md, checkpoint5_list.md, compute_

basis.md) are completely unchanged. They specify the methodology and mathematical formulations; this amendment corrects an implementation variable-name disconnect.

Freeze v5: hashes recorded in paper/proposals/FREEZE.sha256; commit recorded in paper/STATUS.md.

Amendment 5: 10 Nov 2026 (Pre-registered NR-G10: non-oracle residual recipe at the two neighbouring operating cells; before any NR-G10 run) Trigger. The operating-grid evaluation G10 evaluated the M1 recipe, which auto-selected the base rule on 20 calibration traces. In both cells (G2_hi_d10 and G3_lo_d18), the selector chose FixedPU (Oracle) (offset -2 in G2, -3 in G3). G10 yielded no WIN in either cell (Claim A passed in both cells, but Claim B failed against Context table in G2, and against FixedPU and Context table in G3), triggering the pre-registered limitation sentence that the M1 result was observed at the M1 operating point only. Meanwhile, the non-oracle residual recipe (Fixed base with calibrated offset) demonstrated clear superiority over the oracle base in NR1 and was confirmed on fresh traces and seeds in NR2 (+3.35 % over Context-joint-500, hierarchical 95 % CI [+2.22, +4.40] %, 9/10 seeds positive). Because the non-oracle recipe is now the recommended configuration, testing whether the non-oracle recipe generalizes across the operating grid is a natural and necessary pre-registered step.

What is added. NR_G10_operating_grid_prereg.md (new, frozen). 1. **Cells & Recipe:** The non-oracle residual recipe (residual A2C with Fixed base rule, one global calibrated offset) evaluated at the two tested neighbouring cells: - G2_hi_d10 (16,000 bits, 10 TTI, base offset Fixed -2) - G3_lo_d18 (8,000 bits, 18 TTI, base offset Fixed -3) 2. **Calibrations:** Frozen single-cue calibrated comparator tables from G10 (Fixed, FixedPU, Context, OLLA tuned; calibrated on 200 traces per cell; hash-checked). 3. **Training & Seeds:** 5 seeds $\times$ 3,000 episodes per cell. Base seeds: 33042 for G2, 34042 for G3 (seeds 33042+201..205 and 34042+201..205; asserted disjoint from all prior rounds, including G10's 31042 and 32042). Primary scheme: 5-seed mean. Secondary scheme: ensemble. 4. **Evaluation Traces:** 300 evaluation traces per cell (G2: seeds 720001+, CRN 730001+; G3: seeds 740001+, CRN 750001+; identical evaluation set to G10). 5. **Comparators & Decision Rules:** Family of 9 single-cue comparators per cell. Rule R1: Holm over 9 per cell; Claim A, Claim B; cell WIN if primary scheme beats all 9. Rule R2: Hierarchical seed $\times$ trace 95 % CI vs best comparator. 6. **Multiplicity Accounting:** Adds 2 cases (G2_hi_d10_NR, G3_lo_d18_NR), 4 scheme evaluations (primary + ensemble per cell). Total cases in study increase from 74 to 76; scheme evaluations increase from 122 to 126. 7. **Pre-registered Wording:** Explicit wording rows for Both WIN, One WINs, Neither WINs, and unrun.

What is not changed. All prior frozen texts (G6, W1, NR1, G10, A6, NR2, common_rules.md, checkpoint5_list.md, compute_basis.md) maintain their specifications. G10 results stand as reported (M1 recipe fails Claim B at both neighbouring cells). NR-G10 evaluates the non-oracle recipe as a separate, distinct pre-registered stage.

Freeze v6: hashes recorded in paper/proposals/FREEZE.sha256.

Amendment 6: 10 Nov 2026 (Implementation-only: removal of redundant 'RetainData', true in dlgradient; memory optimization; verified bit-identical) Trigger. A working-tree change in Optimized/v17/DeepA2C_Shared_v17.m and Optimized/v17/release/code/DeepA2C_Shared_v17.m removed the redundant 'RetainData', true argument from dlgradient(cLoss, criticNet.Learnables, 'RetainData', true). In MATLAB's automatic differentiation, RetainData preserves intermediate activations on the computation graph for subsequent gradient calls. Because actor loss (aLoss) and critic loss (cLoss) are computed on separate network objects (actorNet and criticNet), aLoss gradients never traverse the critic's computation graph. Removing RetainData frees intermediate memory immediately after critic backpropagation, preventing RAM accumulation and accelerating training.

Verification test (PASS: bit-identical). An equivalence test was run training one M1 seed (seed 25243) for 20 episodes under both the committed HEAD version and the working-tree version on the identical channel traces: - Loss trace max absolute difference: 0.00e+00 (isequal: 1) - Entropy trace max absolute difference: 0.00e+00 (isequal: 1) - Actor weights overall max difference: 0.00e+00 (all 6 weight and bias tensors bit-identical) - Critic weights overall max difference: 0.00e+00 (all 6 weight and bias tensors bit-identical) - Best episode identical: 1 (HEAD=20, WORK=20) - Execution speedup: 20.37 s (working-tree) vs 38.22 s (HEAD) - Timestamp verification: DeepA2C_Shared_v17.m was modified at Tue Nov 10 01:03:25 2026, prior to NR2 training (04:42:51–10:06:07). Thus NR2 was trained with the working-tree version, which is verified bit-identical to HEAD.

Freeze v7: Hashes recorded in paper/proposals/FREEZE.sha256.

NR_G10_operating_grid_prereg.md

SHA-256: bb37226469be55b434fb58600e9fce629bf6ef0d379186512116fe2e99222e08

Pre-registration NR-G10: operating-point check with the non-oracle residual recipe (v17x-nr-grid)

STATUS: FROZEN (Amendment 5, 10 Nov 2026, before any NR-G10 run started) (AMENDMENTS.md). §§1–6 do not change after the freeze-v6 commit recorded in paper/STATUS.md; results go in §8. Common rules: common_rules.md. **Class: confirmatory robustness check.** Uses the frozen v17x code; only stage and case configurations are added.

1. Question Stage G10 evaluated whether the M1 recipe (which auto-selected the class-oracle base FixedPU (Oracle) on 20 calibration traces) generalised to two neighbouring operating points (G2_hi_d10 and G3_lo_d18). In G10, neither cell achieved a WIN (Claim A passed in both cells, but Claim B failed against Context table in G2, and against FixedPU and Context table in G3). However, NR1 and NR2 demonstrated that the non-oracle residual recipe (Fixed base with calibrated offset) is superior to the oracle base, beating all comparators including the combined table robustly across seeds (+3.35 % over Context-joint-500, hierarchical 95 % CI [+2.22, +4.40] %, 9/10 seeds). Does the **recommended non-oracle residual recipe** (Fixed base with the cell's calibrated static offset) generalise to the two neighbouring operating points, against the frozen single-cue comparators re-calibrated for each cell?

2. Cells & Recipe Both cells use the non-oracle residual recipe: RTT 8, 8 processes, pf_avg = 'scheduled', drain, useful bits, and residual A2C with 3000 episodes on a **forced Fixed base rule** (no class oracle) using the per-cell calibrated static offsets frozen from G10:

Cell	Packet bits	Deadline (TTI)	Base Rule Offset	Status
G2_hi_d10	16,000	10	Fixed -2	**tested**: higher load at M1's deadline
G3_lo_d18	8,000	18	Fixed -3	**tested**: lower load with looser deadline
G1_lo_d10	8,000	10	—	future work, decided at Checkpoint 5b
G4_hi_d18	16,000	18	—	future work, decided at Checkpoint 5b

- **RL Training:** 5 seeds per cell, 3000 episodes.
  - Base seeds: 33042 (G2) and 34042 (G3), blocks +201..+205. Disjoint from all prior rounds (v9–v17, G10 seeds 31042/32042, NR2 seeds 25042).
  - Primary scheme: the 5-seed mean. Secondary scheme: ensemble.
- **Evaluation:** 300 traces per cell.
  - Evaluation traces: seeds 720001+ (G2) and 740001+ (G3); CRN at +10000 (730001+ and 750001+). Identical trace sets to G10.
- **Comparator Tables:** Frozen from G10 (Optimized/v17x/Dataset/ckpt/g10/calib_single_*.mat, hash-checked). No re-calibration is performed. Single-cue family of 9 comparators per cell (Baseline, OLLA, OLLA tuned, Multi-OLLA, Fixed, FixedPU, Context table, Bandit global, Bandit delay class).

3. Decision Rules

- **Per cell:** family of **9 single-cue comparators**, Holm over 9 (common_rules.md R1). Claim A and Claim B are reported per cell. A cell is a **WIN** if the primary scheme beats all 9 comparators.
- **Seed-level rule** (common_rules.md R2): per cell, the hierarchical seed $\times$ trace 95 % CI of the margin vs the best of the nine comparators. With 5 seeds it is indicative; a WIN whose lower bound is ≤ 0 is marked "not robust to seed resampling".
- **Cells tested:** 2 (G2_hi_d10, G3_lo_d18).

4. Pre-registered Wording

Outcome	Paper statement
Both cells WIN (2 of 2)	"With the non-oracle residual recipe, the gain held at both tested neighbouring operating points (a higher load at the M1 deadline, and a lower load with a looser deadline): residual A2C on a calibrated static offset beat the OLLA family and the single-cue calibrated rules re-calibrated for each cell (margins +a % and +b % vs the best rule). Two further grid corners were not tested."
One cell WINs (1 of 2)	"With the non-oracle residual recipe, the gain is operating-point dependent: residual A2C on a calibrated static offset beat the OLLA family and the single-cue rules at [cell] but not at [cell] (per-cell table). Two further grid corners were not tested."
No tested cell WINs (0 of 2)	"With the non-oracle residual recipe, the result was observed at the M1 operating point only; at the two tested neighbouring points it was not reproduced (per-cell table). Two further grid corners were not tested."
NR-G10 not run	"The non-oracle residual recipe was confirmed at the M1 operating point only; neighbouring operating points were not tested under this recipe."

5. Compute Basis Per cell (no calibration pass needed; comparators are frozen from G10): - RL training: 5 seeds × 3000 episodes × 0.14 s ≈ 0.58 h. - Evaluation: 300 traces × 23 scheme runs (7 deterministic, 10 bandit, 6 RL) × 0.095 s ≈ 0.18 h. - **Per cell:** ≈ 0.76 h (sequential) / ≈ 0.4 h (`n_workers = 2`). - **Two cells:** ≈ 0.8 h wall time (`n_workers = 2`, basis upper bound ≈ 1.5 h).

6. Multiplicity Adds **2 cases** (`G2_hi_d10_NR`, `G3_lo_d18_NR`) with 2 rows each (primary 5-seed mean and ensemble). Total cases across the project: 74 → **76**. Total scheme-evaluation rows: 122 → **126**.

7. Stage Specification for the Runner (`Optimized/v17x/`, stage `nr_g10`) Implemented as `stage_nr_g10(ctx)`; per cell: 1. Loads frozen comparator calibration from `ckpt/g10/calib_single_<cell>.mat`. 2. Enforces non-oracle base rule: `env.base_rule_kind = 'fixed'`, `env.fixed_offset = -2` (G2) or `-3` (G3). 3. Trains 5 seeds (`train_seed_<cell>_1..5`). 4. Evaluates on 300 traces against the 9 comparators. 5. Outputs: `Dataset/table_cells_nr.csv`, `Dataset/REPORT_nr_g10.txt`.

8. Results *(empty until run)*